

A time-geographic approach to studying tourist-resident interactions in a travel destination

Mengyao Ren^a, Qili Gao^b, Shih-Lung Shaw^c, Sangwon Park^d, Yang Xu^{e,f,g,*}

^a College of Geography and Planning, Chengdu University of Technology, Chengdu, China

^b Shenzhen Audencia Financial Technology Institute, Shenzhen University, Shenzhen, China

^c Department of Geography and Sustainability, University of Tennessee, Knoxville, United States

^d Smart Tourism Education Platform, College of Hotel and Tourism Management, Kyung Hee University, Seoul, Republic of Korea

^e Department of Land Surveying and Geospatial Science, The Hong Kong Polytechnic University, Hong Kong, China

^f Research Centre for Digital Transformation of Tourism, The Hong Kong Polytechnic University, Hong Kong, China

^g Smart Cities Research Institute, The Hong Kong Polytechnic University, Hong Kong, China

ARTICLE INFO

Keywords:

Time geography
Tourist-resident interaction
Spatiotemporal behavior
Activity-based analysis

ABSTRACT

Understanding how tourists and residents interact through everyday travel and activity participation is critical for addressing social tensions in travel destinations. Existing assessments of tourism pressure largely rely on aggregate indicators, such as tourist-to-resident ratios, which overlook how these two groups interact across space, time, and activity venues in their daily behavior and provide limited details for designing nuanced policy interventions. This study addresses this gap by proposing an activity-informed time-geographic framework that conceptualizes tourist-resident interactions as a multidimensional behavioral process unfolding across space, time, and activity participation. Building on classic time-geographic concepts of space–time paths and bundles, the framework identifies four interaction modes and characterizes them into a hierarchical set of co-location scenarios with different levels of direct contact potential. Global and local indices are developed to quantify interaction potential under different scenarios. Using geolocated debit and credit card transaction data from Jeju, Korea, the empirical results demonstrate that commonly used aggregate indicators overestimate tourism pressure by ignoring behavioral segregation between the two groups across space, time, and activity venues. Interaction hotspots are highly concentrated in dining and retail venues during lunch and dinner hours. Spatially, distinct interaction patterns emerge across various areas, like in city centers, tourists and residents engage in different activities simultaneously, whereas in resort areas, they share the same activities but at different times. This study extends time-geographic approaches by incorporating the activity dimension into social interaction analysis. It provides insights into the details of tourist-resident frictions, supporting more targeted and adaptive destination management strategies.

1. Introduction

Understanding how different population groups interact through everyday travel and activity participation is a central concern in contemporary travel behavior research, particularly in destinations facing increasing pressures from tourism growth (Xu et al., 2019; Zhou et al., 2020; Victoriano et al., 2020). In recent years, overtourism has emerged as a critical social challenge in many cities, where the influx of tourists leads to overcrowding in public spaces, disrupting residents' daily lives and elevating social tensions (UNESCO, 2024; Vu et al., 2021;

Cheung and Li, 2019; Sharpley, 2014; Bimonte and Faralla, 2016). While policy interventions such as visitor caps aim to protect local community well-being, they present a profound policy challenge: how to mitigate the friction of overtourism without diminishing the economic benefits. Addressing this challenge requires moving beyond blanket restrictions toward nuanced destination management. However, current research lacks a systematic analytical framework that helps stakeholders logically decode tourist-resident interactions from the global to the detailed local level, making it difficult to assess conflict risks and implement targeted interventions.

* Corresponding author.

E-mail addresses: mengyaoren@cdut.edu.cn (M. Ren), qlgao@szu.edu.cn (Q. Gao), sshaw@utk.edu (S.-L. Shaw), sangwon.park@khu.ac.kr (S. Park), yang.ls.xu@polyu.edu.hk (Y. Xu).

<https://doi.org/10.1016/j.tbs.2026.101345>

Received 30 December 2025; Received in revised form 8 June 2026; Accepted 27 June 2026

Available online 6 July 2026

2214-367X/© 2026 Hong Kong Society for Transportation Studies. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Classically, the *Tourist Intensity (TI)* index is the most widely utilized indicator for quantifying social pressure at travel destinations (Lundberg, 1974; McElroy, 2003). By measuring the ratio of total tourists to the resident population, it offers a straightforward metric to reflect overall tourism pressure and tourism growth. However, in urban contexts, interactions between tourists and residents arise from they share use of urban spaces and public facilities (Snepenger et al., 2003; Sherlock, 2001; Derrett, 2003). Given that shared facilities are typically clustered spatially and subject to specific temporal patterns of use, the interaction potential is usually unevenly distributed across space and time (Belisle and Hoy, 1980; Sheldon and Var, 1984). Thus, the classic aggregate approach has been criticized for its oversimplifying assumption of homogeneous impacts and overlooked internal spatiotemporal heterogeneity (Wall and Mathieson, 2006; Dyer et al., 2007; Mason and Cheyne, 2000; Tosun, 2002).

Recent studies have leveraged spatiotemporal big data (e.g., mobile phone positioning, GPS tracking, and transaction records) to investigate tourist-resident interactions at much finer spatial and temporal resolutions (Li et al., 2018; Su et al., 2022; Mashkov and Shoval, 2023; Chen et al., 2024). These high-resolution studies provide crucial insights by revealing the nuanced behavioral dynamics hidden beneath aggregated totals. For instance, by downscaling observations to fine-grained grids, recent research reveals that tourist activities exhibit extreme spatial concentration rather than even distribution, such as a small number of urban hotspots consistently attract a large proportion of tourist activity (Cheng et al., 2025), creating severe localized pressures that city-wide averages obscure.

While recent studies have made great advances, existing spatial-temporal analyses offer limited discussion on the distinct modes of interactions under different space-time relationships and activity contexts. Tourists and residents usually share urban spaces with different daily routines and activity demands (Belisle and Hoy, 1980; Sheldon and Var, 1984; Li et al., 2018; Chen et al., 2024). Residents are typically bound by fixed functional schedules (e.g., commuting), whereas tourists operate with flexible leisure itineraries. Social friction often emerges when these divergent routines intersect at specific times and venues, such as competing for transit capacity during morning rush hours or sharing local restaurants at dinner. Consequently, intergroup interactions naturally unfold into diverse and complex modes. From the perspective of *Contact Theory* (Allport, 1954), the specific context and nature of an activity fundamentally determine whether an encounter fosters cultural exchange or exacerbates resource competition. Indeed, high-quality, multifaceted interactions encompassing diverse purposes can significantly reduce perceived cultural distance and foster emotional solidarity (Fan et al., 2017; Tan and Hsu, 2024). Despite this, existing spatial-temporal analyses frequently treat physical co-locations as conceptually equivalent, failing to differentiate distinct interaction modes and activity contexts.

To address this limitation, we introduce a time-geographic framework for examining interactions across space and time (Hägerstrand, 1970). Time geography conceptualizes individual movements as space-time paths, and potential interactions occur when these paths converge in shared locations, forming space-time bundles (Miller, 2005b; Neutens et al., 2011; Parkes and Thrift, 1980; Miller, 2004; Shaw, 2012; Sui, 2012). In physical space, these bundles can manifest as either *Synchronous Presence* (co-location in both space and time, enabling direct face-to-face contact) or *Asynchronous Presence* (co-location in space but at different times). In realistic tourism scenarios, *Asynchronous Presence* often occurs through the shared use of physical infrastructure and facilities. For instance, tourists flocking to a local beach or residential park in the afternoon may deplete physical resources or alter the neighborhood's atmosphere, indirectly impacting the experience of residents who visit the same location in the evening. These concepts emphasize that the alignment of temporal trajectories, alongside spatial co-location, jointly shape the nature of interaction and the potential for direct contact (Miller, 2005a; Shoval and Isaacson,

2007).

However, existing *SP* and *AP* concepts do not incorporate the activity dimension, limiting their applicability in scenarios requiring a clear identification of activity venues where these interactions occur. Actually, activity dimension is inherently embedded within the analytical framework of time geography. As Hägerstrand (1970) discussed in the context of coupling constraints, the purpose of an activity fundamentally shapes the nature of the interaction. More broadly, understanding the activities and social contexts in which interactions take place allows us to move beyond absolute space toward mental space to gain deeper insights into human behavior and perception (Shaw and Sui, 2020). Activities determine whether tourists and residents engage in cooperative (e.g., participating in festivals) or competitive (e.g., competing for limited restaurant seats) behaviors, influencing their tolerance for each other and their interaction experiences (Jurowski and Gursoy, 2004; Raymond and Brown, 2007; Liu et al., 2024). Therefore, it is critical to enrich the activity context within the existing time-geographic framework to meet the analytical requirements in contemporary studies.

To systematically capture these multifaceted tourist-resident interactions, this study proposes an activity-informed time-geographic framework. First, it characterizes the four interaction modes between tourists and residents by integrating the activity dimension into existing *Synchronous* and *Asynchronous Presence* concepts. Second, by progressively adding spatial, temporal, and activity dimensions, it conceptualizes a hierarchical set of co-location scenarios. Building upon this hierarchical structure, corresponding global and local indices are developed to quantify interaction potentials. This design aims to bridge global aggregate indices and fine-grained localized indices, yielding a crucial dual advantage: it retains the overall view of tourism pressure provided by traditional indices whilst comprehensively capturing local heterogeneity. Through this hierarchical analysis, we can not only quantify the interaction potential across different scenarios but also reveal more nuanced behavioural characteristics by tracking variations in indices across different levels of analytical dimensions. These variations provide important insights into the mechanisms shaping social friction, including spatial separation, temporal segregation, and activity divergence. Detailed definitions of the conceptual framework and the associated indices are provided in Section 3.

To demonstrate the empirical applicability of the proposed framework, this study uses a geolocated credit and debit card transaction dataset from Jeju, Korea, spanning January 2019 to September 2020, to examine tourist-resident interactions. This dataset records the locations, times, and activity types of consumer spending by tourists and residents. Compared with commonly used location-based service (LBS) data (e.g., GPS trajectories or mobile phone signals), this dataset offers a key advantage in capturing individuals' actual activity participation. Since each transaction reflects a specific activity carried out at a particular venue, the data inherently incorporate the spatial, temporal, and activity dimensions of behavior without requiring additional data sources or indirect inference to reconstruct activity types. Applying the proposed framework to this dataset, the analyses reveal city-level tourist-resident interaction potential, spatial variations of their interaction patterns utilizing data from a pre-pandemic normal period. Furthermore, data from the COVID-19 pandemic, alongside the 2019 tourism peak season, are utilized exclusively in a dedicated section to validate the framework's capability in detecting dynamic behavioral shifts during special periods. These results demonstrate the framework's ability to produce granular insights beyond traditional aggregate metrics.

This study proposes a time-geographic framework that integrates activity types into co-location analysis, enhancing our understanding of interaction patterns between tourists and residents. Beyond tourism, the framework offers a generalizable tool for examining urban social interactions across different population groups, such as different ethnic or income groups. It emphasizes moving beyond traditional social interaction assessments that focus on encounters in space and time, toward understanding the activity contexts in which interactions occur to gain

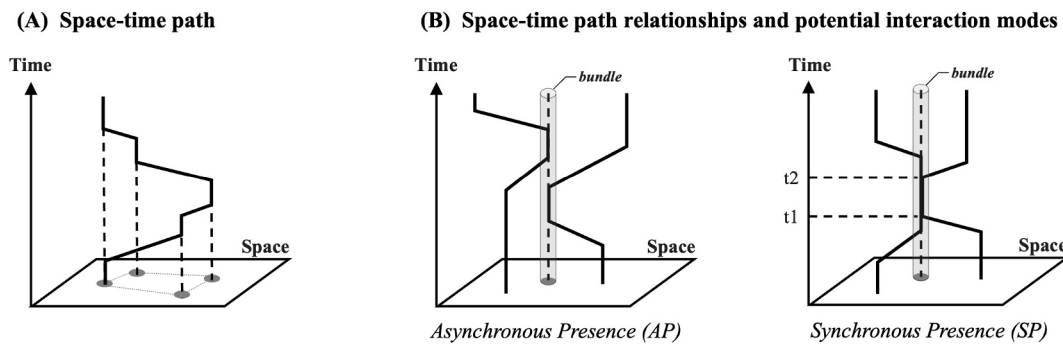


Fig. 1. Examples of space-time path and space-time bundle. (A) Space-time path; (B) Two types of space-time bundles and the corresponding potential interaction modes (adapted from Shaw & Yu, 2009).

more meaningful insights.

2. Literature review

2.1. Social interaction through the lens of time geography

Interactions between individuals largely require physical proximity to enable meaningful engagement. Time geography conceptualizes potential interactions through the relationships between individuals' space-time paths (Hägerstrand, 1970). As illustrated in Fig. 1A, a space-time path traces an individual's sequence of activities across locations over time. Social interactions occur when these paths converge in shared locations, which is described as the space-time bundle. It provides a lens to understand how movements intersect to create opportunities for interaction (Parkes and Thrift, 1980; Miller, 2004). As shown in Fig. 1B, there are two basic types of space-time path relationships that enable interaction in physical space (Shaw and Yu, 2009; Miller, 2005a,b; Parkes and Thrift, 1980; Golledge and Stimson, 1997). The first is *Asynchronous Presence (AP)*, introduced by Janelle (1995), which describes situations where individuals occupy the same location at different times. In such cases, interaction occurs indirectly through shared physical objects or facilities. The second is *Synchronous Presence (SP)*, which requires co-location in both space and time, enabling direct face-to-face interaction and contact.

With the proliferation of location-aware technologies and high-resolution tracking data, a substantial body of work within the fields of GIS and mobility science has extended these classical concepts to analyze social interaction (Calabrese et al., 2013; Liu et al., 2015; Xu et al., 2015; Xu et al., 2019; Miller, 2005a,b; Zhang et al., 2025). Several studies have operationalized the ideas of *Asynchronous* and *Synchronous Presence* to measure dynamic patterns of social interactions based on the spatiotemporal proximity of moving individuals (Yin and Shaw, 2015; Shaw and Yu, 2009; Yang et al., 2016; Xu et al., 2017; Chen et al., 2024). Farber et al. (2015) have developed social interaction measures to evaluate how individuals encounter different social groups during their daily travel patterns. Some recent studies have further explored dynamic co-location networks to quantify how people share urban spaces over time (Xu et al., 2017; Xu et al., 2019; Li et al., 2021). These computational mobility analyses represent a significant advancement, shifting the research focus from static social integration to dynamic, proximity-based interactions in urban activity spaces (Kwan and Schwanen, 2016).

However, existing *SP* and *AP* concepts do not incorporate the activity dimension, limiting their applicability in scenarios requiring a clear identification of activity venues where these interactions occur. While current GIS and computational movement analyses excel at capturing the spatiotemporal proximity of interactions (i.e., *where* and *when* people encounter one another), they largely treat all co-locations as conceptually equivalent, paying limited attention to the specific activity context. As Hägerstrand (1970) discussed in the context of coupling constraints, the purpose of an activity fundamentally shapes the nature

of the interaction. For example, residents and tourists may be synchronously present in a commercial district, but tourists' window-shopping may not substantially impact residents working in nearby offices. Conversely, if both groups converge at a restaurant for dining, the shared demand for limited seating generates competition, leading to a sense of crowding and friction.

Moreover, existing studies rarely examine how interaction estimates systematically change across different levels of analytical dimensions. Currently, studies typically operationalize interaction based on available data, ranging from broad shared spatial presence to fine-grained spatiotemporal encounters. However, these varying resolutions are not merely more or less accurate measurements of the same phenomenon; they capture distinct behavioral meanings. Spatial co-location reflects shared urban space, spatiotemporal co-location captures synchronized presence, and activity-based co-location highlights overlapping behavioral intentions. Therefore, each analytical level reveals a distinct perspective of social interaction, and the transitions between these levels contain substantial analytical value. Much like the Modifiable Areal Unit Problem (MAUP) reveals spatial structures across different scales, tracking interaction estimates across these progressively constrained scenarios unveils underlying forms of segregation. For instance, a sharp decline from spatial to spatiotemporal co-location signals a temporal mismatch between groups, while a further drop under activity constraints indicates diverging activity participation. These variations provide important insights into the mechanisms shaping social friction, including spatial separation, temporal segregation, and activity divergence. However, such comparative and hierarchical perspectives remain largely absent in existing interaction studies.

Therefore, existing research on movement interactions exhibits two primary gaps: a semantic gap that overlooks the specific activity context of physical co-locations, and a methodological gap that ignores the hierarchical analytical value of measurement dimensions. To address these gaps, this study proposes a multidimensional time-geographic framework that explicitly incorporates the activity type alongside spatial and temporal trajectories. By progressively integrating spatial, temporal, and activity dimensions into a hierarchical set of co-location scenarios, this framework transitions from observing mere physical proximity to analyzing the underlying resource competition and behavioral segregation, offering a comprehensive tool for understanding social interactions in contemporary urban settings.

2.2. Understanding tourist-resident interactions

Interactions between tourists and residents have long been a central concern in tourism research, as they significantly influence residents' attitudes toward tourism, shape tourist experiences, and affect the social sustainability of destinations (Mody et al., 2019; Sharpley, 2014). Several models and theories have been developed to explain the complex psychological and socio-cultural dynamics of tourist-resident interactions (Butler, 1980; Doxey, 1975; Lundberg, 1974; Krippendorff,

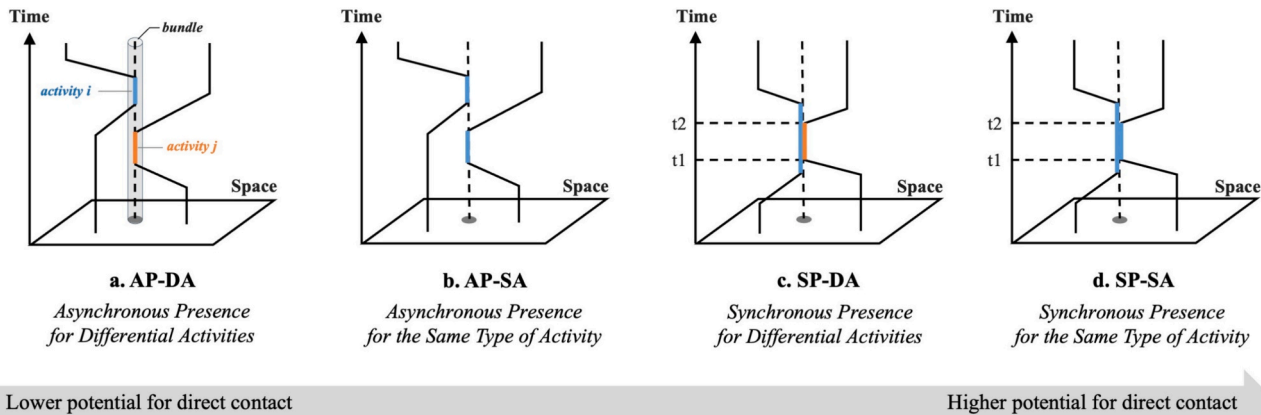
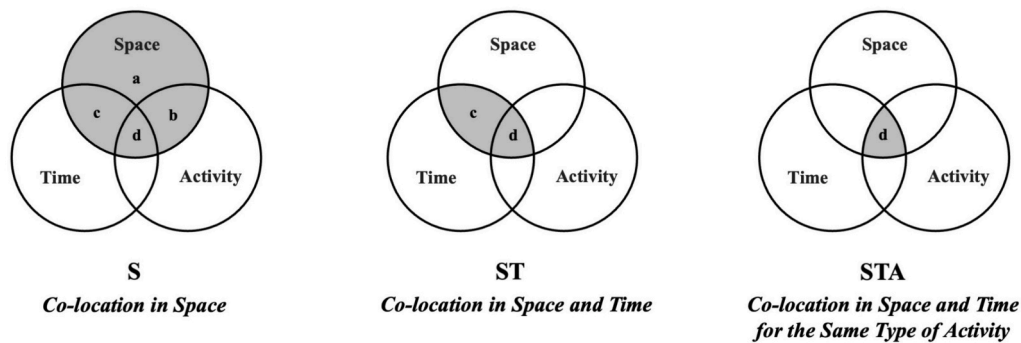
(A) Interaction Modes**(B) Co-location Scenarios**

Fig. 2. Conceptual framework of interaction modes and co-location scenarios.

1987). Doxey's (1975) *Irridex Model* is one of the most influential frameworks, which conceptualizes host community reactions as a progressive shift from euphoria to antagonism driven by an increase in aggregate tourist volumes (Teye et al., 2002). Rooted in this foundational theory, the *Tourist Intensity (TI) index*—calculated as the ratio of tourists to residents—has been widely utilized to assess tourism pressure (Lundberg, 1974; McElroy, 2003; Dumbrovská and Fialová, 2014). The primary advantage of *TI* lies in its simplicity and interpretability, offering a straightforward metric to reflect overall tourism pressure on a destination or the macro-level trends in such pressure as tourism develops. However, this aggregate approach has been criticized for its oversimplifying assumption of homogeneous impacts and overlooked internal spatiotemporal heterogeneity in different urban contexts (Wall and Mathieson, 2006; Dyer et al., 2007; Mason and Cheyne, 2000; Tosun, 2002).

Studies have indicated that the tension relationship between tourists and residents is rarely uniform across a city; it typically emerges in specific locations and time periods (Belisle and Hoy, 1980; Sheldon and Var, 1984). Dense tourist crowds usually concentrate around popular attractions, particularly during peak hours and seasons. Residents living nearby are more likely to perceive overcrowding and suffer the negative impacts of tourism (Jurovski and Gursoy, 2004; Raymond and Brown, 2007; Sharma and Dyer, 2009). In Venice, historic neighbourhoods have seen traditional shops and housing replaced by tourist-oriented businesses, prompting many locals to leave due to rising living costs and a weakened sense of community. Such competition over urban resources between tourists and residents is key to making residents feel an imbalance of power and generating anti-tourism sentiments (Cheong and Miller, 2000; Mody et al., 2019). Therefore, identifying when and where tourist-resident interactions occur is crucial for destination

management and target policy intervention.

With the increasing availability of high-resolution spatiotemporal data (e.g., GPS tracking, mobile phone signaling), a substantial body of tourism research has utilized these emerging datasets to further reveal the dynamic spatiotemporal patterns of how tourists and residents actually co-locate and navigate everyday urban spaces. These studies mark a significant methodological leap by revealing at a fine scale *where* and *when* residents and tourists encounter one another. For instance, scholars have explored how different urban settings shape spatial interactions (Su et al., 2021; Su et al., 2022), dynamic interaction patterns (Chen et al., 2024), and interaction variety within tourism spaces (Tan and Hsu, 2024; Mashkov and Shoval, 2023). While these high-resolution spatiotemporal analyses effectively address the internal heterogeneity that aggregate measures ignore, current literature still exhibits two critical limitations.

First, existing spatial-temporal analyses offer limited discussion on the distinct modes of interactions under different space-time relationships and activity contexts. They frequently treat all physical co-locations as conceptually equivalent, paying limited attention to the different outcomes of contacts. *Contact Theory* (Allport, 1954) emphasizes that the specific context, activity, and nature of intergroup encounters fundamentally determine whether interactions reduce prejudice or foster friction. Guided by such socio-psychological theories, considerable literature has conceptualized tourist-resident interaction as a multidimensional construct rather than a single form of contact. Some studies investigated economic, sociocultural, and environmental dimensions (Andereck et al., 2005; Weaver and Lawton, 2001) and highlighted the critical role of interaction intensity and depth (Pizam et al., 2000). More recently, scholars have explicitly demonstrated that high-quality, multifaceted interactions encompassing diverse purposes and

activities can significantly reduce perceived cultural distance and foster emotional solidarity (Fan et al., 2017). These studies provide a strong theoretical foundation, proving that understanding the specific activity context is important.

Second, current research lacks a computational framework that retains the overall view of tourism pressure provided by traditional indices whilst comprehensively capturing local heterogeneity. Mashkov and Shoval (2023) propose a tourism *Intensity-Density Index (IDI)* at the grid cell level by combining traditional indicators (i.e., tourist-to-resident and tourist-to-area ratios) with high-resolution GPS data. This advancement enabled the mapping of spatial variations in tourism intensity and density across urban spaces, effectively downscaling the physical meaning of aggregate indicators to a fine spatiotemporal granularity. However, different analytical scales tend to answer different questions. Disaggregated high-resolution data are crucial for decoding localized human behaviors, whereas Aggregate indicators like *TI* are essential for reflecting macro trends and ensuring cross-city comparability. Establishing a mathematical connection between global aggregate indicators and fine-grained localized metrics would yield a crucial dual advantage. On the one hand, it would allow researchers and policymakers to leverage aggregate indicators for tracking overarching tourism growth or comparing overall tourism pressure across different cities. On the other hand, it would retain the precision of local-level measurements to pinpoint specific high-pressure areas, venues, or time periods, thereby guiding targeted interventions. However, current research lacks a computational framework capable of bridging these two ends of measurement.

Thereby, by integrating time-geographic principles, this study aims to propose a quantitative framework that explicitly categorizes interaction modes across spatial, temporal, and activity dimensions and captures the interaction potentials across different modes at both global and local levels. In doing so, it preserves the evaluation utility of traditional indices while deeply diagnosing localized tourist-resident interaction contexts, offering a scalable tool for adaptive destination governance.

3. Methodology

3.1. Conceptual framework

This study introduces a conceptual framework that defines four interaction modes between tourists and residents across space, time, and activities. By incorporating activity dimension into the existing concept of the space-time bundle (Fig. 2), it offers a more comprehensive understanding of these interactions.

- a. **Asynchronous Presence for Different Types of Activities (AP-DA)** refers to situations in which individuals visit the same location at different times to engage in different activities. In this context, while there is no face-to-face contact, individuals engage in indirect interactions by sharing the same physical environment. Such indirect interactions can significantly contribute to social tension. For example, the closure of local streets or community venues to prepare for a major tourist event days in advance creates an indirect interaction that disrupts residents' daily functional routines, sparking social friction even without direct encounters.
- b. **Asynchronous Presence for the Same Type of Activity (AP-SA)** refers to situations in which individuals visit the same location at different times to engage in the same type of activity. In this situation, individuals do not have direct contact, but indirect interactions occur through the sequential use of shared facilities and resources. This mode can also exacerbate social tension due to resource depletion or environmental degradation. For instance, tourists flocking to a local beach, residential park, or public parking lot in the afternoon may deplete physical resources or alter the neighborhood's atmosphere, indirectly but profoundly impacting the

experience of residents who visit the same location in the evening for leisure.

- c. **Synchronous Presence for Different Types of Activities (SP-DA)** refers to situations in which individuals visit the same location at the same time to engage in the different types of activities. This mode allows for a higher potential of direct contact. For example, a resident working in a business district may encounter a tourist on the street during lunch hour, creating opportunities for face-to-face interaction or conversation.
- d. **Synchronous Presence for the Same Type of Activity (SP-SA)** refers to situations in which individuals visit the same location at the same time to engage in the same type of activity. This mode carries the highest potential for direct contact, such as dining at the same restaurant or attending the same movie screening. However, it may also increase the risk of competition or conflict between tourists and residents due to overlapping demands on shared spaces and services.

It is important to note that the theoretical combinations of space, time, and activity dimensions can yield eight possible scenarios. However, because this framework is fundamentally rooted in the classic time-geographic concept of the space-time bundle, which necessitates the convergence of space-time paths at a shared physical locale. Thereby, spatial co-location is treated as a strict prerequisite. Consequently, our framework intentionally focuses on the four interaction modes that involve spatial overlap (i.e., co-location in physical space). In this context, these interaction modes represent a gradient of increasing potential for direct contact, beginning with spatial proximity as the foundational requirement, and progressing through temporal synchronization and activity alignment. To facilitate the measurement of interaction potential, this study proposes a hierarchical framework that progressively adds these three dimensions to co-location scenarios, classifying interaction modes into three distinct levels (Fig. 2):

- **Co-location in Space (S)** requires only spatial proximity, regardless of temporal synchronization or activity alignment, encompassing all four modes, i.e., AP-DA, AP-SA, SP-DA, and SP-SA.
- **Co-location in Space and Time (ST)** requires both spatial and temporal coincidence, narrowing the scope of interaction modes to SP-SA and SP-DA.
- **Co-location in Space and Time for the Same Type of activity (STA)** further requires activity alignment, allowing the interaction mode to SP-SA alone.

This framework generalizes the hierarchical relationships of interaction modes through progressively incorporating additional dimensions of co-location. It characterizes which interaction modes are represented by the observed co-location scenarios, thus effectively bridging the concepts of interaction modes to the quantitative measurements. Different scenarios can be analyzed in practical applications depending on the data source and specific requirements. For instance, analyzing co-location in space provides insights into the degree of spatial sharing among different groups. However, such observations often encompass multiple interaction patterns and may overestimate the actual potential for direct contact between groups. Observations of the latter two scenarios offer a more accurate depiction of actual contact potential. By comparing differences across scenarios, the framework also highlights the interaction potential of complementary modes not captured within a given scenario, thus enriching the understanding of intergroup interactions. The framework's flexibility ensures its applicability across datasets with varying information dimensions. It remains functional even when certain dimensions, such as activity types or time, are unavailable, making it a useful tool for diverse analytical contexts.

3.2. Global indices of tourist-resident interactions

Building on the theoretical framework, this study develops indices to

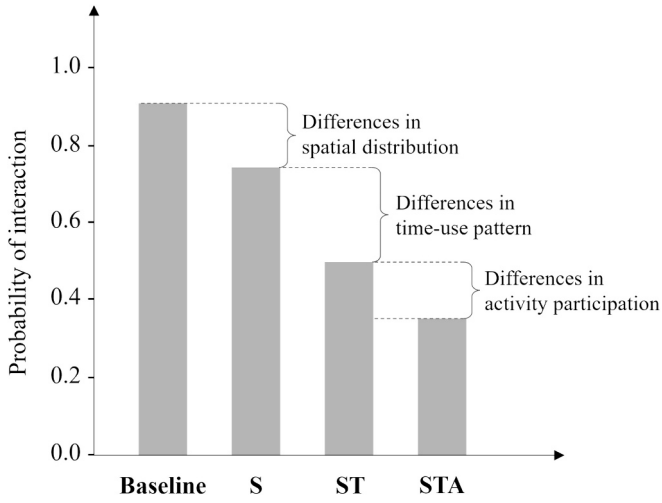


Fig. 3. The relationship between interactions indices and interaction modes in different scenarios.

quantify the potential for tourist-resident interactions in various co-location scenarios. The foundational measurement in our framework is adapted from the classic Exposure Index (also known as the Interaction Index), a widely established metric in social segregation literature (Massey and Denton, 1988; Wong, 2002). Traditionally, the Exposure Index is calculated as the weighted average of the proportion of group x in each spatial unit with the weighting term being the proportion of group y in each unit to the total population of group y . The theoretical maximum of this index equals the overall proportion of group x in the total population, which occurs only when both groups are evenly distributed across all units. In this study, this theoretical maximum value is introduced as a baseline index:

$$\text{Baseline} = X/(X + Y) \quad (1)$$

Here, X and Y denote the total number of recorded activity events (e.g., out-of-home consumptions) generated by tourists and residents across the entire study area, respectively. This definition conceptually aligns with our time-geographic assumption that potential interactions can only occur when individuals are actively participating in activities within shared urban spaces. Accordingly, the Baseline index measures the average probability of tourist-resident interaction under the assumption that these total activities are evenly distributed across the city. It has a mathematical relationship with the classical Tourist Intensity (TI) index, defined as $TI = X/Y$, which can be expressed as $TI = \text{Baseline}/(1 - \text{Baseline})$. The interaction potential between the two groups in the scenario of *Co-location in Space* is quantified by the index S . This index directly applies the aforementioned classic spatial Exposure Index to our context, defined as follows:

$$S = \sum_{i=1}^n \left[\left(\frac{x_i}{x_i + y_i} \right) \left(\frac{y_i}{Y} \right) \right] \quad (2)$$

here, x_i and y_i denote to the total number of recorded activity events generated by tourists and residents in spatial unit i , respectively. n refers to the total number of spatial units in the whole study area. $X = \sum_{i=1}^n x_i$ and $Y = \sum_{i=1}^n y_i$. This index measures the weighted average probability that a resident encounters a tourist in a spatial unit. The theoretical range of S lies between 0 (inclusive) and 1 (exclusive). Assuming that tourists and residents are evenly distributed across all spatial units, the ratio of tourists to residents in each unit i equals the overall ratio for the entire study area. Under this condition, the theoretical upper bound of S is equal to the *Baseline* (for a detailed mathematical derivation, see Appendix A).

To move beyond static spatial sharing and capture the dynamic

nature of human encounters, we developed extensions to this classic spatial metric. Specifically, the formulas for the *Co-location in Space and Time* (ST) index and the *Co-location in Space and Time for the Same Type of Activity* (STA) index were constructed by the authors for this study. By mathematically integrating temporal windows and activity categories into the exposure probability formula, these newly developed indices operationalize the time-geographic concepts of space–time bundles.

The interaction potential between the two groups in the scenario of *Co-location in Space and Time* is quantified by the index ST , defined as follows:

$$ST = \sum_{i=1}^n \sum_{t=1}^m \left[\left(\frac{x_{i,t}}{x_{i,t} + y_{i,t}} \right) \left(\frac{y_{i,t}}{Y} \right) \right] \quad (3)$$

here, $x_{i,t}$ and $y_{i,t}$ denote to the total number of recorded activity events generated by tourists and residents in spatial unit i at time window t , respectively. m refers to the total number of time windows. This index measures the weighted average probability that a resident encounters a tourist in a spatial unit at the same time. Similarly, when the two populations are equally distributed over all time windows, the theoretical upper bound of ST is equal to the S .

The interaction potential between the two groups in the scenario of *Co-location in Space and Time for the Same Type of Activity* is quantified by the index STA , defined as follows:

$$STA = \sum_{i=1}^n \sum_{t=1}^m \sum_{a=1}^w \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] \quad (4)$$

Here, $x_{i,t,a}$ and $y_{i,t,a}$ refer to the total number of recorded activity events generated by tourists and residents in spatial unit i at time window t engaging in activity a , respectively. w refers to the total number of activity types in spatial unit i in which residents and tourists engaged in. This index measures the joint probability of activity co-occurrence—the likelihood that a tourist's activity and a resident's activity overlap in the same space, time, or activity venue, evaluated against the total baseline activity volume. Similarly, when the two groups are equally distributed over all activity venues, the theoretical upper bound of STA is equal to the ST .

Fig. 3 illustrates the relationship among quantitative indices across different scenarios. The scale of each bar represents the probability of interaction between tourists and residents under a given scenario. In a limiting case where tourists and residents are evenly distributed across spatial units, time windows, and activity venues, all three indices reach their theoretical maximums, yielding: $\text{Baseline} = S = ST = STA$ (or a detailed mathematical derivation, see Appendix A). However, in real-world settings, the behavioral patterns of tourists and residents typically differ, leading to uneven distributions across spatial, temporal, and activity dimensions. Therefore, a reasonable hypothesis is: $\text{Baseline} > S > ST > STA$. As shown in Fig. 3, the gap between *Baseline* and S reflects differences in the spatial distribution of the two groups. Conceptually, this mathematical gap inherently captures the total interaction potential of the remaining theoretical combinations where tourists and residents share time or activity types but do not overlap in physical space. This highlights the extent to which traditional tourism indicators may overestimate actual interaction potential. The difference between S and ST captures disparities in time-use patterns among individuals who are spatially co-located. Similarly, the gap between ST and STA reflects differences in activity participation when both groups are co-located in space and time. Collectively, these indices not only quantify the overall interaction potential between tourists and residents under varying co-location conditions but also reveals the heterogeneity of their distribution across space, time, and activity dimensions.

3.3. Decompose global indices by time and activity

In certain circumstances, comprehending the temporal variation of

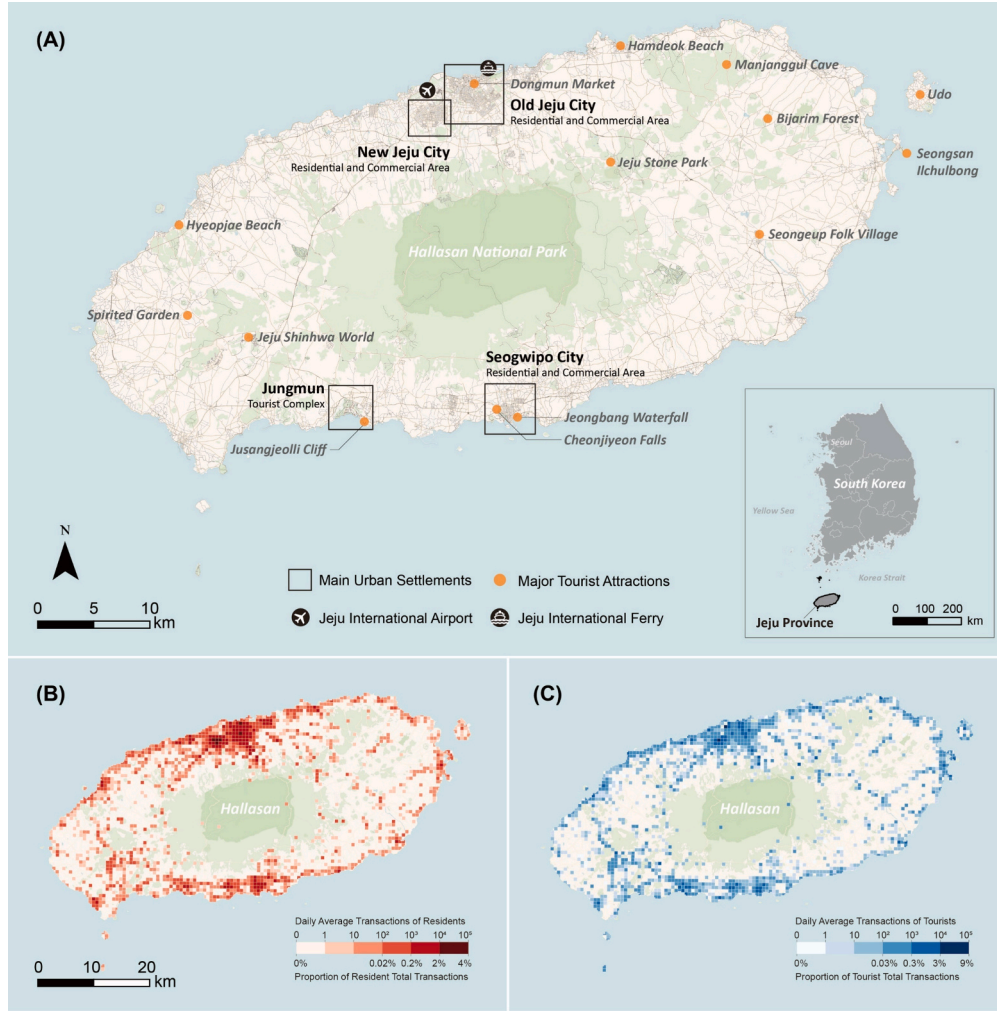


Fig. 4. (A) The location of Jeju in Korea, and spatial distribution of tourist attractions, transport infrastructure and major urban areas in Jeju; (B) Average number of daily transactions by residents on 600 m grid cells; (C) Average number of daily transactions by tourists on 600 m grid cells.

tourist-resident interactions and the differences in interaction potential across activity venues is crucial for destination management. Hence, we introduce time-based ST and STA indices by fixing a time window t :

$$ST_t = \sum_{i=1}^n \left[\left(\frac{x_{i,t}}{x_{i,t} + y_{i,t}} \right) \left(\frac{y_{i,t}}{Y} \right) \right] \quad (5)$$

$$STA_t = \sum_{i=1}^n \sum_{a=1}^w \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] \quad (6)$$

where ST_t and STA_t denote ST and STA interaction potential of tourists and residents during the time period t , respectively. The time-fixed indices enable comparisons between different time periods, thus capturing the temporal trends of the corresponding scenario interaction potentials. The sum of the time-based interaction index for all time periods is equal to the overall interaction index, i.e., $ST = \sum_{t=1}^m ST_t$, and $STA = \sum_{t=1}^m STA_t$.

Similarly, to better understand the differences in interaction potential among activity venues, we introduced activity-based STA by fixing an activity type a . As shown in Equation (7), STA_a denotes STA interaction potential of tourists and residents at activity type a . The sum of the activity-based interaction index for all activity types is equal to the overall interaction index, i.e., $STA = \sum_{a=1}^w STA_a$.

$$STA_a = \sum_{i=1}^n \sum_{t=1}^m \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] \quad (7)$$

We further decompose the STA index by fixing both time t and activity type a to obtain the time-activity-based STA , which could depict the temporal change in interaction potential at specific activity venue (Equation (8)). The sum of $STA_{t,a}$ for all activity types and time periods is equal to the overall interaction index, i.e., $STA = \sum_{t=1}^m \sum_{a=1}^w STA_{t,a}$.

$$STA_{t,a} = \sum_{i=1}^n \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] \quad (8)$$

It is important to note that in Equations (5)–(8), the global denominator Y is consistently utilized rather than a time-specific (Y_t) or activity-specific (Y_a) denominator. This mathematical design is intentional and theoretically grounded. First, it ensures that the calculated indices (STA_t , STA_a , $STA_{t,a}$) represent the joint probability over the entire space–time–activity domain, allowing for direct magnitude comparisons across different time periods and activities. Using a time-specific denominator would artificially inflate the interaction potential during periods with low overall activity but high homogeneity (e.g., nighttime). Second, from a time-geographic perspective, space and time form an integrated activity arena where individuals flexibly allocate and transfer their activities across different temporal and spatial units. Maintaining the global denominator Y allows the framework to capture how

Table 1

Example of transaction records in the dataset.

Date	Time	Grid ID	Spending venue	User type	Number of transactions	Total Expenditure (USD)
2019-01-01	6-9	1****5	general retail	Tourist	5	70.23
2019-01-01	6-9	1****5	general retail	Resident	10	282.34
...	...	...	...	...	...	...
2019-01-02	12-15	9****2	transportation	Tourist	5	564.67
2019-01-03	18-21	9****2	accommodation	Resident	10	1058.76

residents and tourists behaviorally shift and substitute their activity timing and locations throughout the day, a dynamic that would be obscured if observations were isolated into independent time slices. By analyzing the interaction indices, we can gain deeper insights into how the interaction potential between different groups varies across time and activity venues. These findings can inform policymakers and stakeholders, supporting more effective and evidence-based decision-making.

3.4. Local indices of tourist-resident interactions

To identify varying co-location patterns between tourists and residents across different urban environments, we further introduce local level indices. For each spatial unit i , we derive three indices S_i , ST_i , and STA_i , corresponding to the interaction potential in the three co-location scenarios:

$$S_i = \left(\frac{x_i}{x_i + y_i} \right) \left(\frac{y_i}{Y} \right) \quad (9)$$

$$ST_i = \sum_{t=1}^m \left[\left(\frac{x_{i,t}}{x_{i,t} + y_{i,t}} \right) \left(\frac{y_{i,t}}{Y} \right) \right] \quad (10)$$

$$STA_i = \sum_{t=1}^m \sum_{a=1}^w \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] \quad (11)$$

By conducting local-level co-location analysis, we can compare inter-group co-location patterns across diverse urban environments. This analytical approach offers valuable insights into informing tourism management and urban governance strategies.

4. Study area and dataset

To demonstrate the effectiveness of the proposed framework, we conduct an empirical study using a geolocated credit and debit card transaction dataset from Jeju Special Self-Governing Province in the Republic of Korea (hereafter Jeju, Korea). Located in the southwestern part of the country, Jeju comprises Jeju Island and several subsidiary islands, covering a total area of 1847.2 km². As shown in Fig. 4, Jeju Island contains two main cities: Jeju City in the north and Seogwipo City in the south. The central urban areas of these cities serve as the primary residential and commercial centers. Additionally, numerous small settlements, residential zones, and a rich array of tourism facilities and attractions are located along the coastline. Before the COVID-19 pandemic, Jeju received approximately 250,000 tourist arrivals per week (89% domestic), alongside a resident population of around 670,000. The number of tourists and residents makes Jeju an ideal case study for examining tourist-resident interactions.

The geolocated credit and debit card transaction data were provided by South Korea's largest card company, holding a dominant 21.8% national market share. The dataset covers the period from January 1, 2019, to September 30, 2020. To protect privacy, all records were fully anonymized and pre-aggregated by the data provider based on user type, consumption location, time window, and specific venue categories. Users were classified as residents or tourists based on their registration information: those registered in Jeju were identified as residents, while users registered elsewhere in South Korea were categorized as tourists

(domestic inbound visitors only). As shown in Table 1, transactions were aggregated by date, 3-hour time interval, 600-meter grid cell, spending venue, and user type, capturing both the number of transactions and total expenditures. The resulting filtered dataset encompasses over 300 million transactions, averaging approximately 470,000 transactions per day. Within this dataset, transactions made by residents account for roughly 60%, while those by tourists account for about 40%. Fig. 4 B and C present the spatial distribution of average daily transactions by residents and tourists across the grid cells.

This dataset holds significant advantages in this research context, as it captures actual activity participation through recorded consumer behavior. Each transaction reflects a user's card usage at a commercial establishment, providing detailed information on the time of purchase and merchant details, including geographic location and business type. As a result, this dataset inherently incorporates the spatial, temporal, and activity dimensions of behavior, eliminating the need for integrating additional data sources or applying indirect inference to reconstruct activities. However, this dataset inherently excludes activities that do not involve financial transactions (e.g., visiting a public park or a free beach, or simply being present without purchasing). Consequently, spatial units without consumption venues are not captured, which may underestimate the interaction potential in non-commercial public spaces. Despite this limitation, transaction data effectively confirms meaningful, deeper engagement in specific activities, making it highly suitable for demonstrating the proposed activity-informed analytical framework.

Moreover, transaction data enables the differentiation between residents and tourists by comparing the location of the transaction with the user's registered place of residence. User types were predetermined by the data provider. Individuals with a registered address in Jeju were categorized as residents, while those from other regions of Korea were labeled as tourists. Thus, the dataset primarily reflects domestic travelers, who constituted the vast majority (about 89%) of the total 15.29 million visitors of Jeju (Jeju Tourism Organization, 2019), providing a highly representative sample to validate the proposed framework.

The type of spending venue is used as a proxy for activity type in this study. We analyze 11 activity types derived from an initial set of 22 broad categories and more than 1,500 subcategories. To better capture spending behaviors of both residents and tourists, we selected household- and tourism-related categories and reclassified them into 11 types: food and beverage retail, general retail, restaurants, drinking places, transportation, accommodation, indoor recreation, outdoor recreation, health, in-person service, and education. Together, these categories account for over 90% of all transactions. Further details and examples for each category are provided in Appendix B Table 2.

In the empirical analysis, data were aggregated over selected periods to enhance reliability and address the issue of data sparsity. A normal period, from April 1 to May 31, 2019 (61 days), was first selected. During this period, consumption behavior of residents and tourists was relatively stable and did not coincide with the peak tourist season. The data were divided into eight 3-hour intervals across weekdays and weekends, yielding 16 distinct time windows. All records from this period were then aggregated by calculating the average values across 600-meter spatial units, 16 time windows, and 11 types of activity categories. Spatial units were excluded if the average value of any activity type in any time window was less than 1. After this filtering process, a

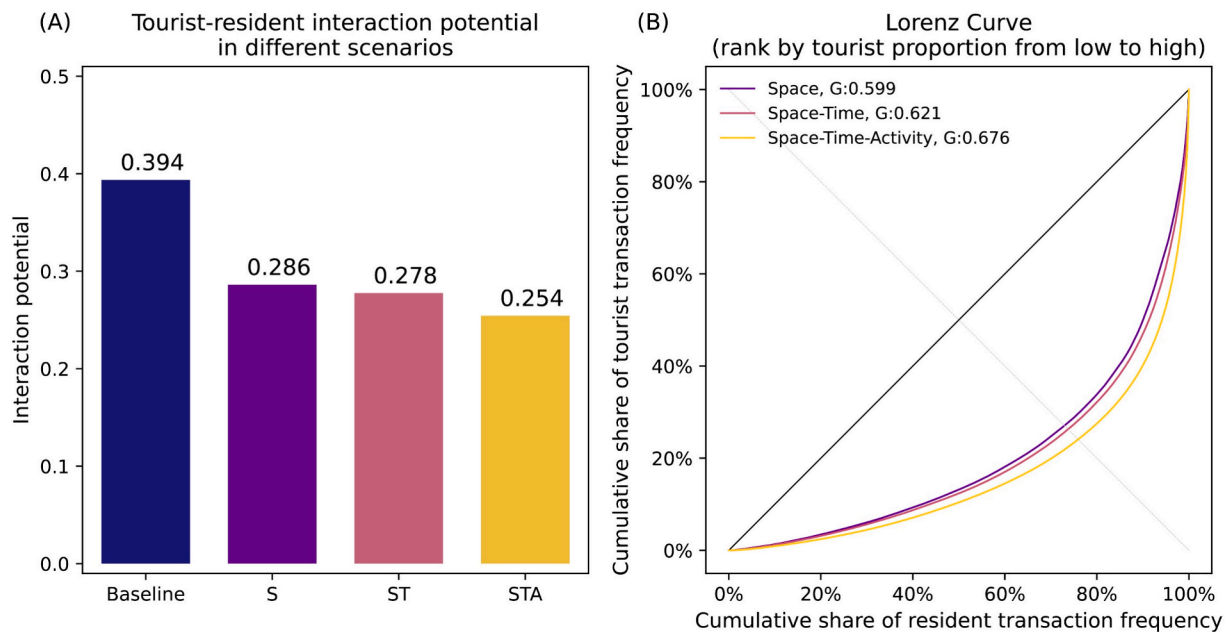


Fig. 5. Tourist-resident interaction potential in different scenarios at the city level. (A) the estimates of the Baseline and interaction indices in different co-location scenarios; (B) Lorenz curves corresponding to the three co-location scenarios and the estimates of the Gini index.

total of 1,502 valid spatial units were retained for analysis. The normal period dataset is used in subsections 5.1, 5.2, and 5.3. In subsection 5.4, the same processing criteria was applied to four special periods for comparative analysis of changes in interactions: the 2019 tourism peak season (July 1–August 31), the first wave of COVID-19 (February 19–April 12, 2020), the stable period (April 12–August 10, 2020), and the second wave (August 11–September 30, 2020). For the three pandemic periods, the corresponding periods in 2019 (same calendar dates) are used as baselines to enable meaningful comparisons. For the tourism peak season, the normal period dataset serves as the comparative baseline.

5. Results

5.1. Overall characteristics of tourist-resident interaction potential

We first estimated the overall tourist-resident interaction potential in Jeju during the normal period using the global indices (Equations 1–4). As shown in Fig. 5A, the value of the *Baseline* index is 0.394. Under the assumption that tourists and residents are evenly distributed across space, this value represents the average probability that a resident encounters a tourist in the city. When *Co-location in Space* is considered, the interaction potential (*S*) decreases to 0.289, reflecting the cumulative weighted probability of a resident encountering a tourist within a spatial unit. The lower value of *S* compared with the *Baseline* suggests that the two groups tend to visit different locations, thereby reducing the overall probability of encountering each other in urban spaces. This result validates that traditional indicators based on aggregate population ratios (e.g., Tourist Intensity) may overestimate actual interaction potential. Further incorporating temporal co-location reduces the interaction potential to 0.278 (*ST*); and incorporating activity type reduces it to 0.254 (*STA*). Overall, the results support the hypothesis of $Baseline > S > ST > STA$, suggesting that differences in spatial distribution, time-use patterns, and activity participation between tourists and residents jointly contribute to progressively lower levels of interaction potential within the city.

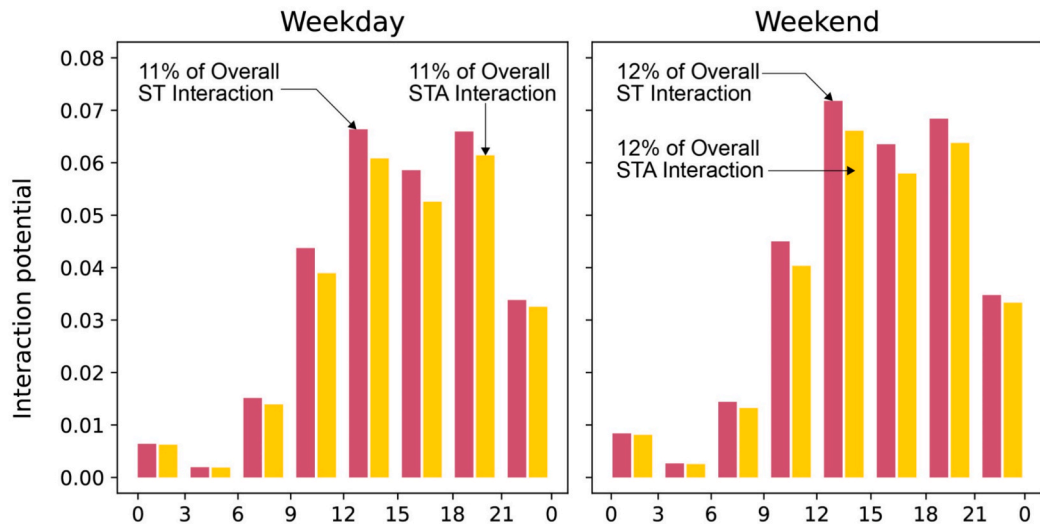
Beyond quantifying the interaction potential under different scenarios, the specific declining trend of these indices reveals city-specific behavioral patterns in Jeju. The significant drop from *Baseline* to *S* and

the minimal gaps between *S*, *ST*, and *STA* reveal that behavioral divergence is predominantly spatial. This leads to a deep insight that in a nature-based destination like Jeju, tourists tend to spend daytime visiting dispersed natural attractions located away from residential areas, resulting in limited spatial overlap with residents. Spatial co-location increases primarily in the evening, when tourists return to urban centers for dining, shopping, and accommodation—activities that residents are also likely to engage in during these periods. Consequently, once they do co-locate in space (typically in urban centers during the evening), they are highly synchronized in both time and activity. This pattern may differ markedly in dense urban tourism destinations such as New York or Hong Kong, where tourists and residents frequently share central urban spaces throughout the day but engage in more divergent activities. This underscores the importance of urban spatial structure and environmental context in shaping tourist-resident interaction patterns. Therefore, the group of indices can serve as a signature of the tourist-resident relationship in each city, enabling a universal framework for cross-city comparisons.

To verify the effectiveness of the proposed indices in capturing behavioral heterogeneity among the two groups, we introduce the Lorenz curve and Gini coefficient for assessment (Fig. 5B). The Lorenz curve illustrates the degree of uneven distribution of tourists and residents across spatial, temporal, and activity dimensions, with curves closer to the diagonal indicating more even distributions (James and Taeuber, 1985). The Gini coefficient quantifies this inequality, calculated as the area between the diagonal and the Lorenz curve (see Appendix C, Equation 20). The results indicate that most of the segregation between tourists and residents stems from differences in their spatial distribution. The dark purple line shows a substantial deviation from the diagonal, with a Gini coefficient of 0.599, indicating that tourists and residents occupy markedly different areas in Jeju. In contrast, adding temporal and activity dimensions produces only modest increases in deviation, with Gini values of 0.621 and 0.676, respectively. This suggests that tourists and residents differ far less in their time-use patterns and activity participation than in their spatial locations. Overall, these indices not only quantify interaction potential but also reveal underlying behavioral differences between the two groups across multiple dimensions.

(A) Temporal Changes in ST and STA Interactions

ST
STA



(B) Percentage of STA interactions at different activity venues

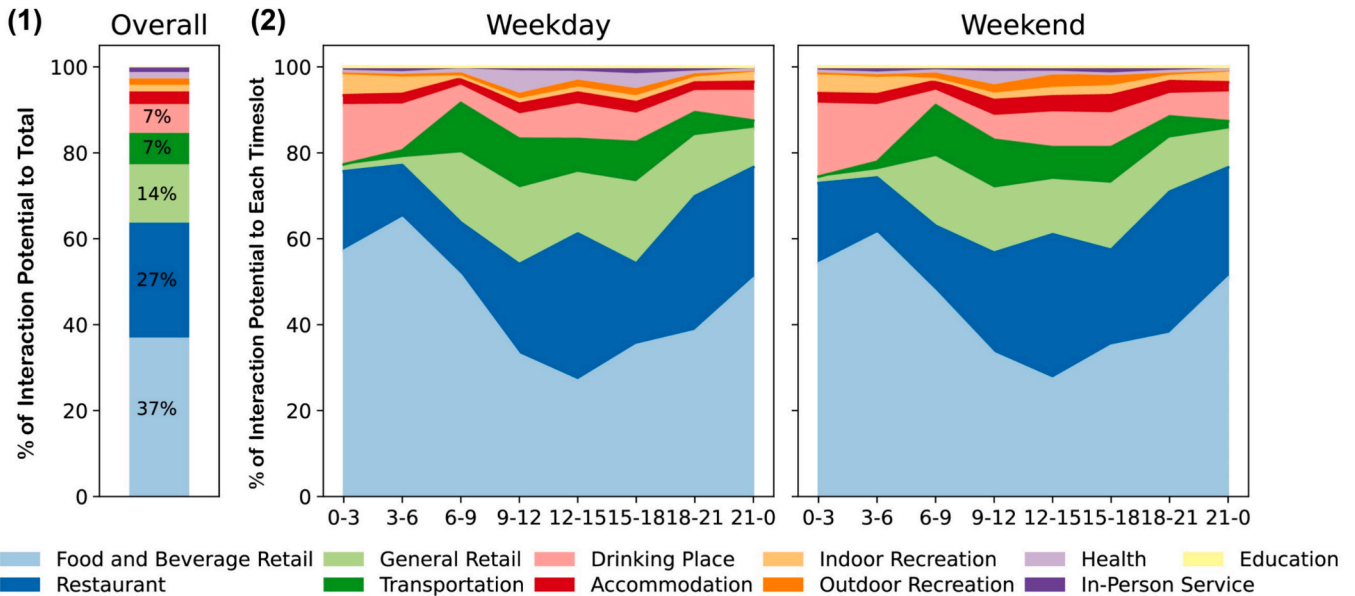


Fig. 6. (A) Variations in tourist-resident interaction potential throughout the day on weekdays and weekends. (B) Distribution of STA interaction potential across activity venues: (1) Percentage of activity-based STA interaction potential relative to overall STA. (2) Percentage of activity-based STA interaction potential relative to STA interaction potential within each time window.

5.2. Temporal and activity-based variations in overall interaction potential

This subsection demonstrates the effectiveness of the proposed framework in capturing temporal and activity-based variations in interaction potential. Fig. 6A presents the time-based ST and STA index values across 16 time windows, including both weekdays and weekends. The results indicate that the interaction potential between tourists and residents is slightly higher on weekends than on weekdays. Within a day, the period from 12:00 to 21:00 exhibits higher interaction potential, accounting for over 60% of the total, with lunchtime (12:00–15:00) and dinnertime (18:00–21:00) representing peak interaction periods. During these periods, STA values are significantly lower than ST values, indicating that although tourists and residents may be co-located in space and time, they are often engaged in different activities, thereby reducing the potential for direct contact.

Fig. 6B illustrates how STA interaction potential is distributed across different activity venues and how this distribution varies over time.

Fig. 6B (1) presents interaction potential by activity type, calculated using the activity-based STA index and expressed as a proportion of the total STA. The results indicate that tourists and residents are most likely to be synchronously present in *Food and Beverage Retail* venues, which account for 37% of the total interaction potential under the STA scenario and therefore present the highest likelihood of direct contact. *Restaurants* follow at 27%, while *General Retail* contributes 14%, and *Transportation* and *Drinking Place* each account for 7%. *Accommodation* as well as *Indoor and Outdoor Recreation* exhibit substantially lower potential for direct contact between tourists and residents.

Fig. 6B (2) illustrates the temporal variation of STA interaction potential across different activity venues, calculated as the proportion of interaction potential for each activity type within each time window relative to the total STA interaction potential during that period. The results indicate that during daytime hours, tourists and residents are most likely to interact directly in *Restaurants*, *General Retail*, and *Transportation* venues. During nighttime hours, the potential for direct contact is higher in *Drinking Places* and *Food and Beverage Retail*. These

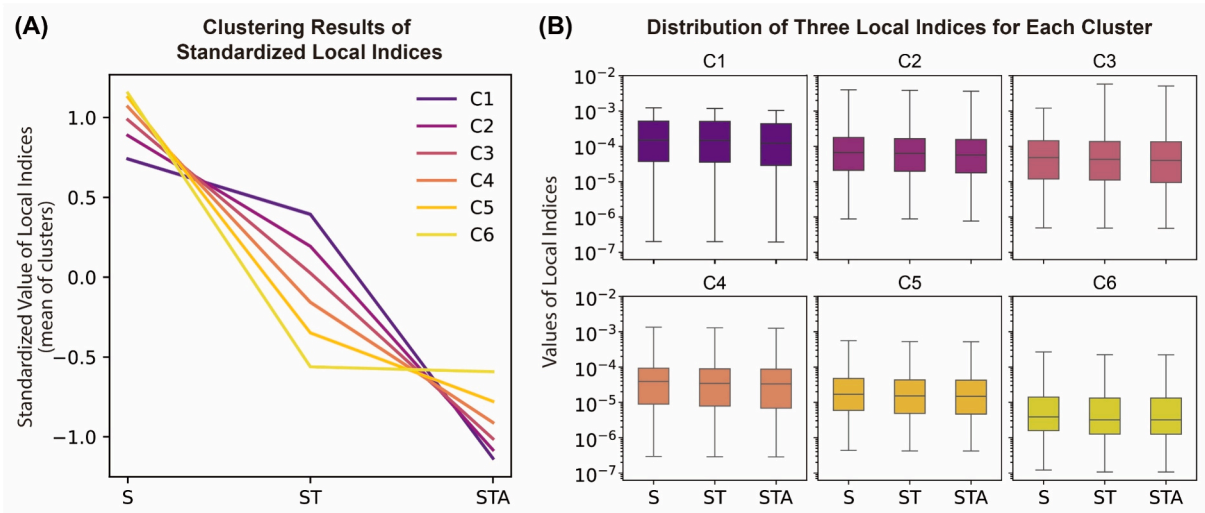


Fig. 7. Results of hierarchical agglomerative clustering. (A) Mean standardized local indices for each cluster, y-axis show the z-score standardized value of local indices; (B) Distribution of local indices across clusters.

patterns suggest that management strategies can be tailored by venue and time period. For instance, during a pandemic, restaurants could implement separate seating for residents and tourists during daytime to reduce transmission risk, while nighttime interventions could focus primarily on drinking places.

5.3. Spatial variations of tourist-resident interactions

Interaction patterns between tourists and residents may vary across urban spaces (Su et al., 2021; Su et al., 2022). To explore these patterns, we applied a hierarchical agglomerative clustering algorithm across spatial units (1,502 in total) (Lukasová, 1979). We used the three local indices (S_i , ST_i , STA_i) as input features, resulting in a 1502×3 matrix. To emphasize the relative behavioral profiles within each spatial unit rather than the absolute magnitude of interaction potential, each row of the matrix (three local indices for each unit) was z-score normalized. This row-wise standardization ensures that the clustering algorithm identifies similarities in the structural dynamics of interactions (e.g., the relative drop-offs from S to ST to STA) rather than clustering units merely by their overall tourist and resident volumes. Initially, each spatial unit was considered an individual cluster. At each iteration, the closest clusters were merged with one another based on the proximity matrix measured by Euclidean distance. We utilized Ward's linkage method in order to minimize within-group variance while maximizing variance among the groups. The optimal number of clusters was determined by evaluating the corresponding dendrogram (hierarchical tree diagram) based on the similarities and dissimilarities of the spatial units (Appendix D Fig. 11). By identifying the cut-off threshold where merging clusters further would result in a significant jump in linkage distance (i.e., within-cluster variance), and to satisfy the condition of balanced cluster sizes, six distinct clusters were selected.

Fig. 7A presents the interaction patterns of the six clusters, represented by the mean standardized values of the three local indices for each cluster. From C1 to C6, the gap between S and ST increases, while the gap between ST and STA decreases. This indicates that tourists and residents display increasing divergence in time-use patterns but progressively more similar participation in activity types from C1 to C6. For example, in Cluster C1, tourists and residents follow similar temporal routines yet engage in different activities. C2 experiences a moderate drop-off from S to ST but a substantial decline to STA, indicating that tourists and residents share time but engage in divergent activities. Conversely, C3 and C5 exhibit much steeper declines from S to ST, revealing highly asynchronous visitation patterns where the two groups

rarely occupy the space simultaneously. In Cluster C6, tourists and residents participate in similar activities but follow different temporal routines.

Fig. 7B shows the distribution of the absolute values of the three local indices across clusters, revealing a progressive decline in median interaction potential from C1 to C6. Interpreted together with the interaction patterns in Fig. 7A, clusters with higher interaction potential (e.g., C1 and C2) tend to feature a diverse mix of activity venues, which enhances their attractiveness but also leads to greater heterogeneity in activity participation between tourists and residents. In contrast, clusters with lower interaction potential (e.g., C5 and C6) are typically dominated by fewer activity types, attracting a narrower range of users and resulting in lower interaction potential but higher similarity in activity participation between the two groups.

Fig. 8 shows the spatial distribution of the six clusters and the proportion of spending by residents and tourists in different activity locations within each cluster. The proportion of spatial units in each cluster to the total number of spatial units is as follows: 19.9% (C1), 13.3% (C2), 11.2% (C3), 14.0% (C4), 7.8% (C5), and 33.9% (C6). Spatial units in clusters C1 and C2 are mainly concentrated in urban centers and small suburban settlements (e.g., development zones and residential neighborhoods). In these areas, resident consumption accounts for a larger share than tourist consumption, indicating a resident-oriented urban environment with spending concentrated in restaurants, bars, and retails. The potential for interaction between tourists and residents is relatively high in these areas, as they tend to be present at the same time while engaging in different types of activities. Spatial units in C3 are near beaches, transportation hubs (e.g., airports), and.

tourist resorts. Here, tourist spending significantly surpasses that of residents, reflecting the dominance of tourism-oriented activities. Cluster C4 comprises small residential areas that also contain tourist hotspots, exhibiting a mixed-use character. In these areas, tourists' and residents' spending patterns are similar, indicating balanced and integrated interaction patterns. Clusters C5 and C6 are characterized by single-function tourism areas, such as theme parks and remote islands. Tourist spending predominates, particularly in indoor and outdoor recreations. These areas exhibit lower interaction potential, as tourists and residents typically visit at different time periods. Overall, these results highlight the critical role of urban spatial structure and activity venue diversity in shaping tourist-resident interaction patterns.

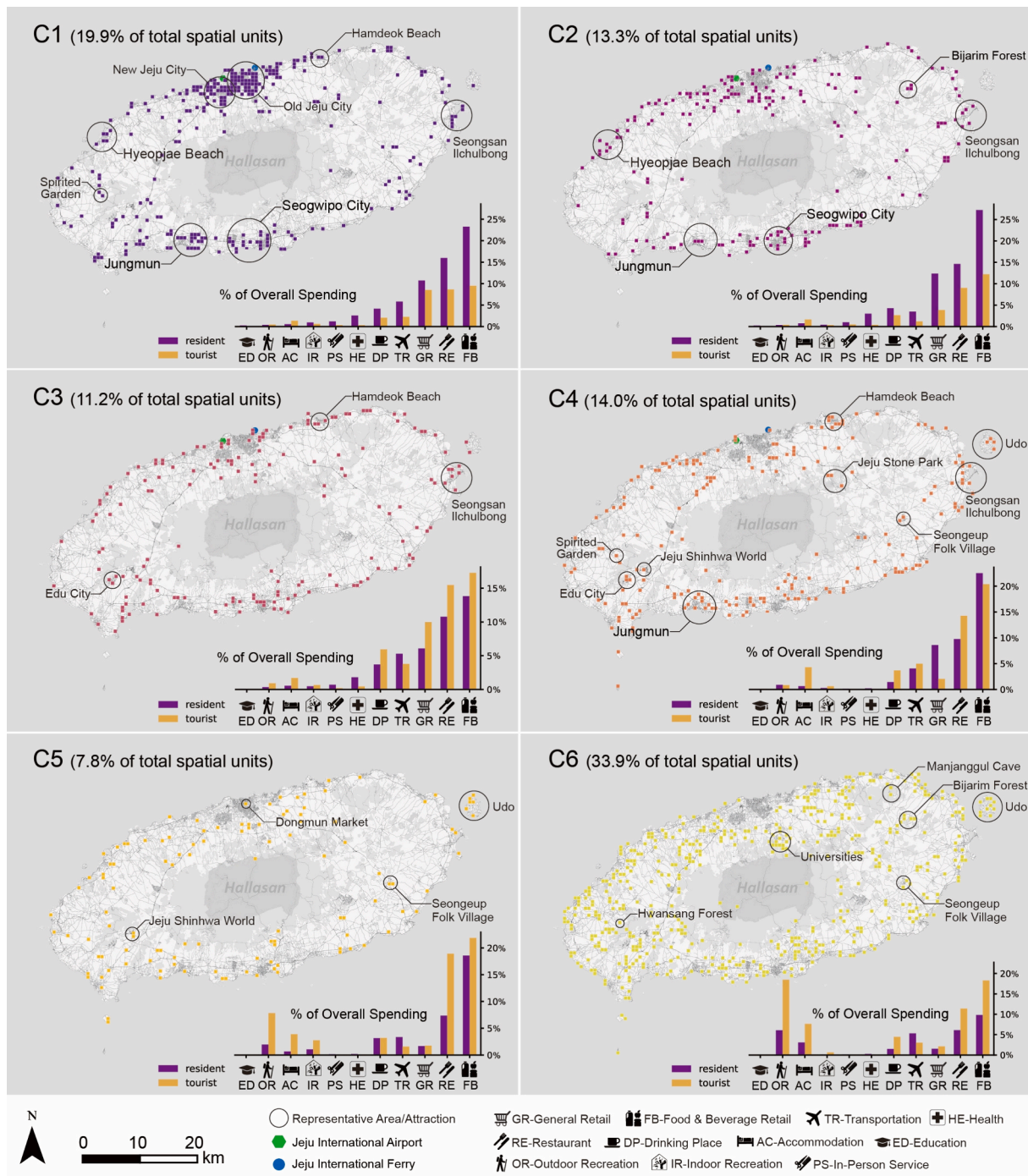


Fig. 8. Spatial distribution of clusters and the proportion of consumer spending allocated to different activity venues within each cluster.

5.4. Changes in tourist-resident interaction potential during special periods

This subsection illustrates how the proposed framework can help identify and understand changes in underlying behavioral patterns during special periods. Four periods were analyzed: the 2019 tourism peak (2019-07-01 to 2019-08-31), the first wave of the COVID-19 outbreak in 2020 (2020-02-19 to 2020-04-12), the stable period (2020-04-12 to 2020-08-10), and the second wave (2020-08-11 to 2020-09-30). Each period was compared with a corresponding baseline period: April–May 2019 for the tourism peak, and the same timeframe in 2019 for the other three periods. Overall interaction potential at the city level was calculated for each period and its baseline period, and

percentage changes were then computed to quantify shifts in interaction potential.

Fig. 9A shows that tourist-resident interaction potential increases across all scenarios during the peak tourist season. However, the increases in the S , ST , and STA indices are smaller than that of the *Baseline* index, indicating that interaction potential is not only driven by the number of tourists. This difference can be attributed to the spatial concentration of tourists at popular attractions during peak season. Such concentration increases spatial heterogeneity between tourists and residents, limiting the growth of the S index. In contrast, the ST and STA indices increase more than the S index, suggesting that temporal and activity-based differences between tourists and residents have narrowed. During the pandemic, interaction potential decreased

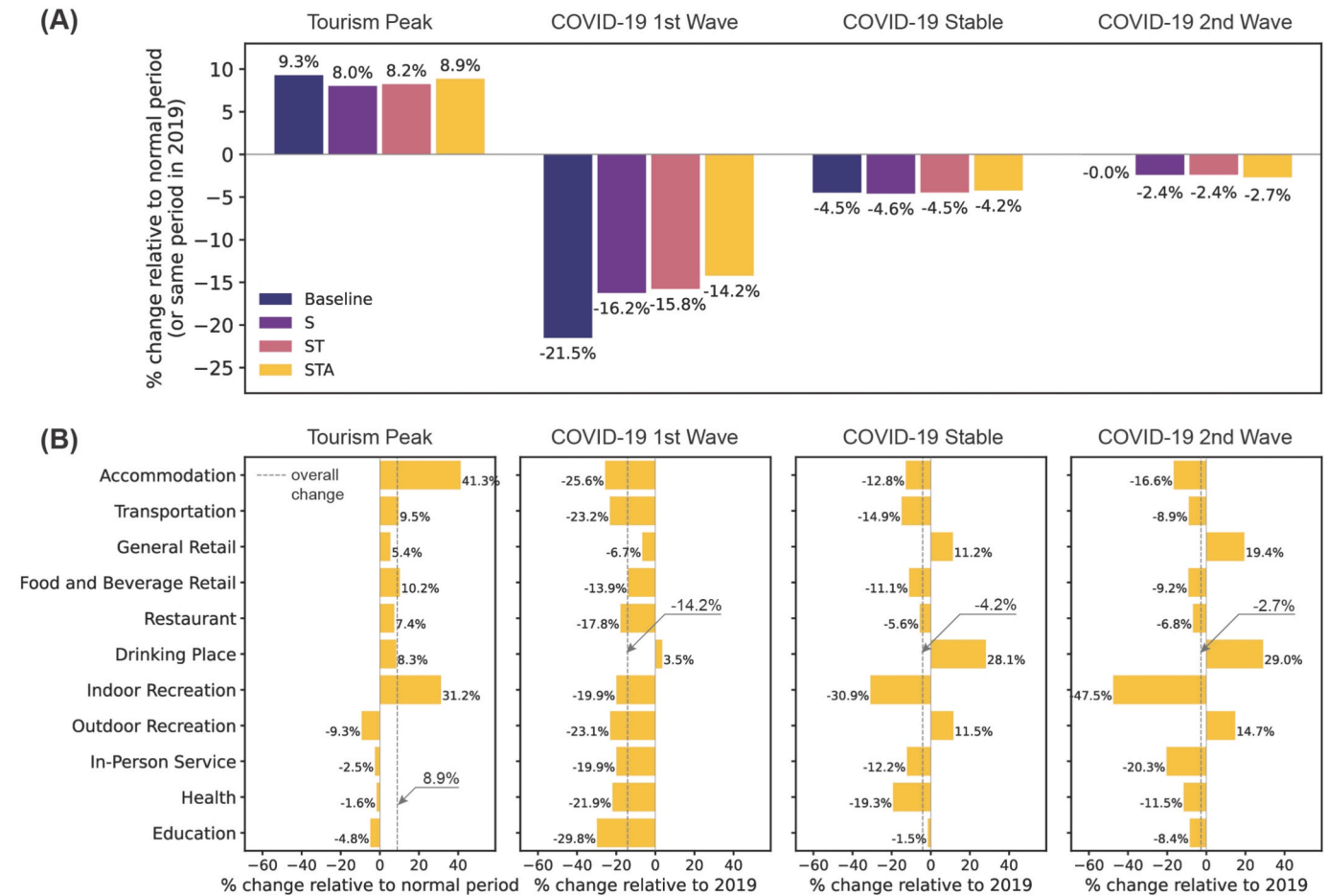


Fig. 9. Changes in tourist-resident interaction potential during the four special periods relative to their corresponding base periods. (A) Changes in overall interaction potential for different co-location scenarios; (B) Changes in interaction potential in different activity venues.

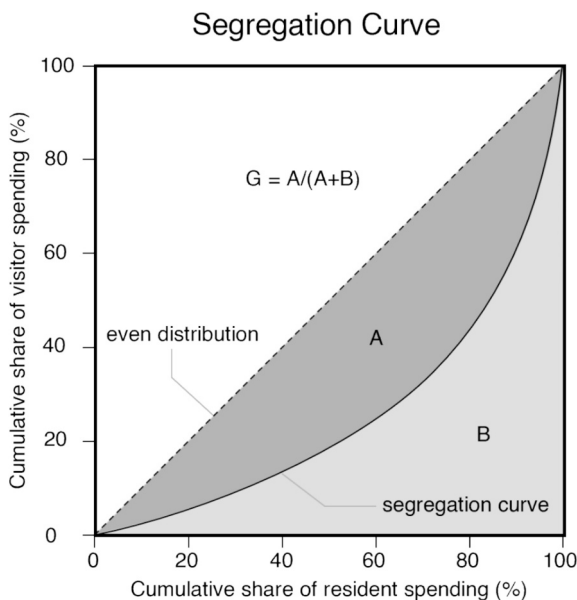


Fig. 10. Lorenz curve and Gini index.

substantially across all scenarios, with the sharpest decline occurring during the first wave. Although the reduction eased during the stable period and the second wave, the interaction indices remained lower than

the *Baseline* decreases. This indicates that, beyond the decline in tourist numbers, heightened spatial, temporal, and activity-related segregation further suppressed the potential for interaction between these two groups.

We next examined changes in the potential for direct contact between tourists and residents in different activity venues using the activity-based *STA* index (Fig. 9B). During the peak tourist season, interaction potential increased sharply in accommodation and indoor recreations but decreased in other venues, like outdoor recreations. This suggests that the challenge of heightened tourist-resident interaction potential during peak seasons does not affect all activity venues. During the pandemic, interaction potential declined substantially across all venues during the first wave. In the subsequent stable period and second wave, trends varied by venue: interaction potential continued to decrease in restaurants, food and beverage retail, and indoor recreations, but steadily increased in general retail, drinking places, and outdoor recreations. These patterns align with previous findings that people favored lower-risk outdoor activities and reduced dining out during the pandemic (Ren et al., 2024). These results suggest that rather than applying blanket restrictions, targeted management of high-contact-risk venues can effectively reduce conflicts and health risks while allowing safer activities to continue, thereby balancing public health concerns with the social and economic needs of both tourists and residents. In summary, the analysis in this section goes beyond simply capturing temporal variations. By analyzing the gaps between these indices, the framework reveals how tourists and residents adapt to external shocks. It clearly shows whether their behaviors converge or diverge across space, time, and activities.

6. Conclusion and discussion

6.1. Conclusion

This study addresses the critical challenge of social tension in tourism destinations by proposing an activity-informed time-geographic framework to examine tourist-resident interactions. Moving beyond the traditional host-guest theoretical model (e.g., *Doxey's Irridex Model*), the proposed framework conceptualizes interactions as a multidimensional behavioral process unfolding across space, time, and activity venues through systematically decomposing intergroup co-location into a hierarchical set of scenarios. By developing a group of interaction indices across scenarios at both global and local levels, this study effectively bridges the gap between aggregate tourism pressure and granular localized behavioral dynamics. Therefore, the proposed framework can not only quantify the interaction potential under different scenarios, but also help researchers logically investigate and better understand behavioral patterns under a space-time-activity context.

The empirical application using large-scale transaction data in Jeju yields three core takeaways. First, at the overall level, the absolute values and specific declining trend of the indices serve as a signature of overarching behavioral patterns: tourists and residents remain largely segregated most of the time, but exhibit highly synchronized temporal and activity patterns (e.g., dining) when they eventually co-locate in shared urban spaces (e.g., commercial areas). This reveals a typical pattern for destinations where attractions and residential areas are spatially separated (e.g., natural-based destinations). Second, at the local level, the framework uncovers complex spatial heterogeneity, demonstrating that interaction patterns vary across different urban settings (e.g., from activity-driven segregation in mixed-use urban centers to time-driven segregation in single-function resorts). Third, by tracking changes in indices over a specific period (e.g., the COVID-19 pandemic and peak seasons), this framework successfully captures how changes in interaction potential are shaped by the combined shifts in visitor numbers and their behavior patterns across space, time, and activity venues. These empirical analyses demonstrate how the proposed framework enables us to progressively deconstruct the interaction potential and behavioural patterns between tourists and residents, moving from the global to the local level. The framework's sensitivity to external shocks and specific events demonstrates its generalizability to examine human behavioral shifts under various extreme urban scenarios.

6.2. Theoretical contributions

This study advances tourism and urban geography by providing a quantitative analytical framework that deeply integrates time-geographic principles with intergroup interaction theories. First, this study successfully incorporates activity dimension into concepts of interaction modes, thereby bridging the semantic gap in proximity-based interaction research. While previous spatiotemporal analyses excel at capturing physical co-locations, they largely treat all encounters as conceptually equivalent, overlooking the specific interaction modes and activity context. By identifying four distinct interaction modes (i.e., *AP-DA*, *AP-SA*, *SP-DA*, and *SP-SA*) based on spatial, temporal, and activity alignments, this framework explicitly addresses the calls from socio-psychological literature. Theories such as *Contact Theory* (Allport, 1954) and recent studies emphasizing interaction variety and multifaceted contacts (Fan et al., 2017; Tan and Hsu, 2024) argue that the nature and type of an activity fundamentally determine whether an encounter fosters cooperation or conflict. By differentiating these diverse interaction types and operationalizing them into a hierarchical set of co-location scenarios, this study provides a crucial computational bridge that links abstract theoretical discussions of multidimensional social dynamics with quantifiable, data-driven spatial measurements.

Second, this study contributes a methodological innovation by introducing a hierarchical analytical perspective to multidimensional

interaction measurements. While previous methodological advancements have successfully downscaled aggregate metrics to high-resolution grids, they typically evaluate interaction potential at static time slices and compare them across different temporal granularities. These analyses tend to treat human mobility as spatial displacements within fixed periods, overlooking the fact that human activities continuously flow and transition across the spatiotemporal domain. Consequently, relying on static time slices based observation can mask nuanced time-allocation behaviors and underlying activity rhythms. The proposed framework drives a paradigm shift by structurally extending classic social segregation indices (Massey and Denton, 1988) into progressive co-location scenarios (*S*, *ST*, *STA*). Rather than treating measurement dimensions merely as static filters or boundaries, this novel perspective fully leverages the hierarchical analytical value among them. By leveraging the magnitude of the gaps between these indices, the framework can uncover the underlying mechanisms of social friction (i.e., spatial avoidance, temporal mismatch, and activity divergence), transforming the measurement from simply observing physical proximity to systematically decoding behavioral pattern.

6.3. Practical implications

From a practical perspective, this framework provides an actionable analytical tool for stakeholders, enabling a shift from aggregate volume control (e.g., blanket visitor caps) to evidence-based adaptive urban governance through step-by-step downscaling analysis. For instance, in single-function resort areas characterized by time-driven segregation, managers should focus on temporal staggering strategies (such as implementing time-slot reservations or off-peak pricing for residents) to optimize the capacity of shared facilities without severely restricting tourist volumes. Conversely, in dense urban centers where tourists and residents share space and time but engage in different activities, urban planners should prioritize mixed-use spatial zoning to physically buffer residents' commuting and daily routines from tourist sightseeing interference. Furthermore, the framework's sensitivity to external shocks supports dynamic crisis management. By identifying shifts in contact potential across specific activity venues during events like the COVID-19 pandemic or peak tourism seasons, authorities can adjust service capacities at high-risk, shared-activity venues (e.g., dining and indoor recreation) while allowing lower-risk spatial co-locations to continue, thereby balancing public health and social well-being with economic resilience.

The proposed framework provides a systematic analytical tool that fundamentally extends the traditional measurement of tourism pressure allowing for cross-city comparisons. Traditionally, destination managers rely on the Tourist Intensity (TI) index (the ratio of tourists to residents) to assess social carrying capacity. However, a critical limitation of TI is its lack of cross-city comparability, as there is no universal ideal ratio. For instance, in 2019, the TI in Hong Kong was approximately 6:1, whereas in Macau it reached an overwhelming 51:1. Relying solely on TI would suggest that Macau faces far more severe social tension. Yet, Hong Kong is a high-density urban destination where residents' living spaces serve as the most popular tourist attractions. Conversely, Macau is renowned for its casino resorts, where tourist activities and residential neighborhoods are largely separated in space, yielding limited substantive intergroup interaction. Therefore, our proposed framework effectively addresses this limitation. By mathematically establishing a direct link between the TI and our *Baseline* index (i.e., $TI = \text{Baseline} / (1 - \text{Baseline})$), this framework serves as an advanced, multi-dimensional extension of TI. By progressively incorporating spatial, temporal, and activity heterogeneity, the resulting *S*, *ST*, and *STA* indices measure an interaction potential that is much closer to reality. Consequently, this framework can extend traditional TI applications, providing a standardized and reliable basis for cross-city comparisons regardless of absolute visitor and resident volumes.

6.4. Limitations and future work

We would like to point out a few limitations about the geo-located transaction dataset. (1) While the massive transaction volume ensures market representativeness, the pre-aggregation prevents tracking the exact sample size of unique individuals (i.e., the exact sample size of tourists and residents). To ensure the dataset robustly supports our analysis, we verified its validity and representativeness by comparing the trend of tourists' daily transaction volumes in our dataset with the official statistics of total tourist arrivals to Jeju (provided by the Jeju Special Self-Governing Province Tourism Association). The highly consistent alignment between these two trends indirectly but strongly demonstrates that our dataset accurately reflects the overall human activity dynamics and variations on the island.

(2) The dataset does not include international arrivals. Because international tourists often exhibit highly concentrated spatial behaviors (e.g., clustering around iconic landmarks or duty-free shops), this bias might lead to underestimate the degree of spatial segregation and the overall interaction potential. Nevertheless, domestic tourists constituted the majority of visitors to Jeju, approximately 89% in 2019; the outbreak of the COVID-19 pandemic and the subsequent international travel restrictions further decimated international arrivals, resulting in an even higher proportion of domestic visitors. This ensures that our findings capture the most representative portion of tourism activities. Future research could incorporate international roaming data to explicitly compare the distinct interaction patterns of domestic and international groups with local residents.

(3) Interactions in non-consumption environments are not recorded. This data bias may artificially narrow the observed gaps between the *S*, *ST*, and *STA* indices. To address this, future studies should fuse transaction records with continuous trajectory data (e.g., GPS or mobile signaling) to reconstruct a more comprehensive spectrum of actual tourist-resident encounters.

(4) As the data are aggregated, the spatial and temporal scales may influence the measurement of interaction potential due to the Modifiable Areal Unit Problem (MAUP). While coarser spatial or temporal units may obscure underlying spatial heterogeneity and result in overestimating co-location, excessively fine units can lead to severe data sparsity (e.g., zero transactions in many micro-cells), rendering the calculation of interaction indices unstable and statistically meaningless. Limited by data availability, this study employs the finest resolution available in the dataset (600 m and 3-hour intervals), which provides a robust and practical foundation for our framework but captures only the potential exposure of individuals rather than providing a direct measure of interaction. Future studies could refine activity classifications or incorporate emerging data sources (e.g., GPS or mobile phone data) at multiple scales to further validate the optimal resolution for measuring interaction potential.

(5) Regarding the dataset time frame, this study utilizes transaction records spanning from 2019 to 2020. Given that post-pandemic spatial-temporal behaviors may have shifted, the specific empirical interaction patterns observed in Jeju during this period may differ from current realities. Specifically, the empirical results derived from this timeframe captured unique behavioral adaptations driven by acute health concerns, such as a sharp decline in indoor recreation and heightened temporal and activity segregation. If measured today, as global travel normalizes and pandemic-induced anxieties recede, we might observe a rebound in synchronous interactions within dense

indoor venues and more intertwined tourist-resident space-time paths.

However, it is important to note that this study is not a purely empirical study. A key contribution of this paper is the theoretical insights and analytical capabilities gained from the proposed activity-informed time-geographic framework. The proposed framework is intentionally flexible and data adaptive. Although it integrates spatial, temporal, and activity dimensions, it can be applied using only those dimensions available in a given data context. But the interpretation is grounded in the specific characteristics of the city. In nature-based destinations such as Jeju, tourist-resident segregation is primarily spatial, with limited temporal and activity differentiation; thus, spatial co-location alone may offer a reasonable proxy for assessing interaction potential when finer-grained data are unavailable. By contrast, in dense and multifunctional urban destinations (e.g., Hong Kong, New York, Paris), tourists and residents often share space and time but engage in different activities, making it necessary to consider all three dimensions to accurately assess interaction potential.

Additionally, this framework has a clear spatial implication. It does not capture resource competition without direct spatial overlap. For example, tourists boarding upstream might exhaust public transport capacity, leaving no room for residents downstream. Evaluating these system-wide conflicts is also crucial for understanding overall tourism impacts. Future research could integrate continuous tracking data with network-based mobility models to examine those potential interactions and further expand the framework's analytical boundaries.

CRedit authorship contribution statement

Mengyao Ren: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Qili Gao:** Writing – review & editing. **Shih-Lung Shaw:** Writing – review & editing, Conceptualization. **Sangwon Park:** Writing – review & editing, Data curation. **Yang Xu:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini to check the grammar and improve its clarity and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank the editor and anonymous reviewers for their valuable comments that improved this article. This research is jointly supported by the Research Centre for Digital Transformation of Tourism (Grant No. 1-BBFG) and Otto Poon Charitable Foundation Smart Cities Research Institute (Grant No. J-CDD4), The Hong Kong Polytechnic University.

Appendices.

A. Theoretical maximum of the global indices

Assume that tourists and residents are evenly distributed across all spatial units. Then, for each spatial unit *i*, the ratio of tourists to residents is

equal to the overall ratio in the study area, i.e.,

$$\frac{x_i}{y_i} = \frac{X}{Y} \quad (12)$$

where x_i and y_i denote the number of tourists and residents in unit i , and X and Y are the total numbers of tourists and residents in the entire study area. Let, $k = \frac{X}{Y}$, so that:

$$x_i = k \bullet y_i \quad (13)$$

Substituting into the expression for the interaction share in unit i :

$$\frac{x_i}{x_i + y_i} = \frac{k \bullet y_i}{k \bullet y_i + y_i} = \frac{k}{k + 1} = \frac{X}{X + Y} \quad (14)$$

Thus, the interaction index S becomes:

$$S = \sum_{i=1}^n \left[\left(\frac{X}{X + Y} \right) \left(\frac{y_i}{Y} \right) \right] = \frac{X}{X + Y} \bullet \frac{\sum_{i=1}^n y_i}{Y} = \frac{X}{X + Y} \quad (15)$$

Therefore, the theoretical upper bound of S is:

$$S = \frac{X}{X + Y} = \text{Baseline} \quad (16)$$

Similarly, when the two populations are equally distributed over all time windows, we have $\frac{x_{i,t}}{y_{i,t}} = \frac{x_i}{y_i}$ for each time window t within spatial unit i . Therefore, like the equation 4.13 and 4.14, we have $\frac{x_{i,t}}{x_{i,t} + y_{i,t}} = \frac{x_i}{x_i + y_i}$. Thus, the interaction index ST becomes:

$$ST = \sum_{i=1}^n \sum_{t=1}^m \left[\left(\frac{x_{i,t}}{x_{i,t} + y_{i,t}} \right) \left(\frac{y_{i,t}}{Y} \right) \right] = \sum_{i=1}^n \sum_{t=1}^m \left[\left(\frac{x_i}{x_i + y_i} \right) \left(\frac{y_i}{Y} \right) \right] = S \quad (17)$$

When the two populations are equally distributed over all activity venues, we have $\frac{x_{i,t,a}}{y_{i,t,a}} = \frac{x_{i,t}}{y_{i,t}}$ and $\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} = \frac{x_{i,t}}{x_{i,t} + y_{i,t}}$. Thus, the interaction index STA becomes:

$$STA = \sum_{i=1}^n \sum_{t=1}^m \sum_{a=1}^w \left[\left(\frac{x_{i,t,a}}{x_{i,t,a} + y_{i,t,a}} \right) \left(\frac{y_{i,t,a}}{Y} \right) \right] = \sum_{i=1}^n \sum_{t=1}^m \left[\left(\frac{x_{i,t}}{x_{i,t} + y_{i,t}} \right) \left(\frac{y_{i,t}}{Y} \right) \right] = ST \quad (18)$$

Therefore, in a limiting case where tourists and residents are evenly distributed across spatial units, time windows, and activity venues, all three indices reach their theoretical maximums, yielding:

$$\text{Baseline} = S = ST = STA \quad (19)$$

B. Reclassified consumption categories

Table 2

Reclassified consumption categories and the percentage of each category in total expenditure and transactions.

Category	Example of Transaction Types	No. of subtypes	Resident Transactions	Resident Expenditure	Tourist Transactions	Tourist Expenditure
Food and Beverage	Convenience stores, supermarkets, retail & wholesale of food and beverage, e.g., fruit and vegetables, meat, fish and marine products, dairy products, rice cakes, etc.	83	35.35%	18.80%	27.28%	11.72%
Retail						
Restaurant	Korean style restaurants, confectioner shops, pizza, hamburger, sandwich, noodle houses, bars and canteens, chicken shops, lunch counters, western style restaurants, etc.	16	21.48%	15.96%	23.93%	21.40%
General Retail	Retail & wholesale: clothing, cosmetics and perfumery, gifts, novelties and souvenirs, etc.	350	16.43%	28.88%	16.00%	20.01%
Transportation	Automotive gas/oil stations, renting of motor vehicles, coastal water passenger transport, vehicle parking facilities, urban transit systems, charter bus transport, etc.	49	8.06%	13.11%	5.41%	6.09%
Drinking Place	Non-alcoholic beverages places, other drinking places, general amusement drinking places	6	6.63%	5.05%	7.39%	3.69%
Health	General clinics, general hospitals, dental clinics, oriental medical clinics, child day care services, para-medical services, etc.	18	4.11%	6.85%	0.75%	0.97%
In-Person Service	Personal care services: hair beauty, saunas, skin beauty, etc. Household services: household laundry services, repair of household machinery, etc.	43	1.49%	1.97%	0.64%	0.67%

(continued on next page)

Table 2 (continued)

Category	Example of Transaction Types	No. of subtypes	Resident Transactions	Resident Expenditure	Tourist Transactions	Tourist Expenditure
Indoor Recreation	Computer game room, singing room, museum, billiard room, bowling alley, swimming pool, library, reading room, physical fitness facility, etc.	23	1.13%	0.63%	1.36%	0.85%
Outdoor Recreation	Golf and skiing facilities, amusement and theme park, botanical and zoological gardens, natural parks, etc.	23	1.05%	1.73%	2.71%	3.65%
Accommodation	Hotels, inns, condominium, juvenile camps, renting of non-residential buildings, etc.	23	1.03%	1.42%	4.54%	8.58%
Education	General subject educational institute, private educational institute, fine arts schools, foreign language schools, sports education, universities, other technical and vocational secondary education, driving schools, etc.	25	0.36%	2.33%	0.08%	0.32%
Total			97.12%	96.73%	90.10%	77.96%

C. Lorenz curve and measures for evenness

To construct a Lorenz curve of tourists' and residents' spending, plot the cumulative proportion of tourist transactions as a function of the cumulative proportion of resident transactions after sorting observed units (i.e., spatial units, space-time units, or space-time-activity units) into descending order according to the percentage of resident transactions in the total transactions of the unit (Fig. 10). The diagonal line indicates the condition of even distribution. The line for a completely segregated condition would lie along the x-axis from 0 to 1 and then rise along the y-axis. As such, the closer the curves are to the diagonal line, the more even the distribution of tourists and residents is across the observed units; conversely, the more separated they are.

The Gini index is one of the most used indicators of the evenness of social segregation, which has a determinate relationship to the Lorenz curve. It is always equal to the area between the diagonal and the Lorenz curve expressed as a fraction of the total area below the diagonal, i.e., $A/(A + B)$ in Fig. 10. It is calculated as the mean absolute difference between minority proportions weighted across all pairs of areal units, expressed as a proportion of the maximum weighted mean difference (Massey and Denton, 1988). The formula is as follows:

$$G = \frac{\sum_i \sum_j t_i t_j |p_i - p_j|}{2T^2 P(1 - P)} \quad (20)$$

where t_i and t_j denote the total number of transactions in units i and j , respectively. p_i and p_j denote the proportion of resident transactions in units i and j , respectively. T denotes the total number of transactions for both tourists and residents across all units. P denotes the proportion of resident transactions to the total transactions. G equals zero only if $p_i = P$ for all i , and equals 1 only if each cell contains a single type of population. Thus, G varies between 0 and 1 for the no-segregation and completely segregated conditions, respectively. This study uses the Lorenz curve and the Gini index to assess the uneven distribution of tourists' and residents' spending in the three scenarios.

D. Hierarchical clustering tree diagram

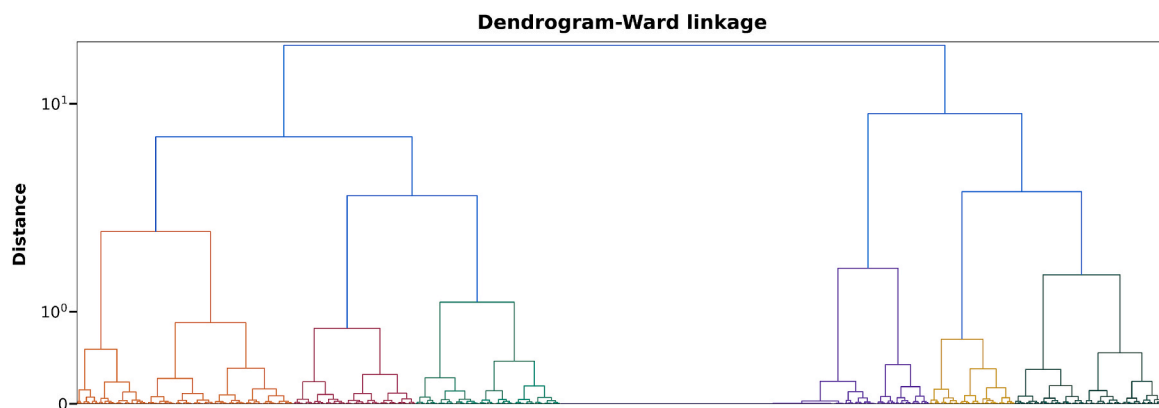


Fig. 11. Hierarchical clustering tree diagram by Ward's linkage methods. We select the Ward's linkage method to minimize a within-group variance while maximizing between-group variance.

References

- Andereck, K.L., Valentine, K.M., Knopf, R.C., Vogt, C.A., 2005. Residents' perceptions of community tourism impacts. *Ann. Tour. Res.* 32 (4), 1056–1076.
- Allport, G.W., 1954. *The Nature of Prejudice*. Reading/Addison-Wesley.
- Belisle, F.J., Hoy, D.R., 1980. The perceived impact of tourism by residents a case study in Santa Marta, Colombia. *Ann. Tour. Res.* 7 (1), 83–101.
- Bimonte, S., Faralla, V., 2016. Does residents' perceived life satisfaction vary with tourist season? A two-step survey in a Mediterranean destination. *Tour. Manag.* 55, 199–208.
- Butler, R.W., 1980. The concept of a tourist area cycle of evolution: Implications for management of resources. *Can. Geogr.* 24 (1), 5–12.
- Calabrese, F., Diao, M., Di Lorenzo, G., Ferreira Jr, J., Ratti, C., 2013. Understanding individual mobility patterns from urban sensing data: a mobile phone trace example. *Transp. Res. Part C Emerging Technol.* 26, 301–313.

- Cheung, K.S., Li, L.H., 2019. Understanding visitor-resident relations in overtourism: developing resilience for sustainable tourism. *J. Sustain. Tour.* 27 (8), 1197–1216.
- Chen, Z., Yang, J., Liu, X., Guo, Z., 2024. Reinterpreting activity space in tourism by mapping tourist-resident interactions in populated cities. *Tour. Recreat. Res.* 49 (4), 797–811.
- Cheng, J., Xu, Y., Jian, I.Y., Ren, M., Park, S., 2025. Spatial concentration of intra-urban tourist activities and inter-group differences between Asian, European and north American travelers in Korean cities. *Tour. Manag.* 107, 105064.
- Cheong, S.M., Miller, M.L., 2000. Power and tourism: a Foucauldian observation. *Ann. Tour. Res.* 27 (2), 371–390.
- Derrett, R., 2003. Making sense of how festivals demonstrate a community's sense of place. *Event Management* 8 (1), 49–58.
- Doxey, G. V., 1975. A causation theory of visitor-resident irritants: Methodology and research inferences. In: *Conference proceedings of the travel research association. San Diego, p. 195e198.*
- Dumbrovská, V., Fialová, D., 2014. Tourist intensity in capital cities in Central Europe: comparative analysis of tourism in Prague, Vienna and Budapest. *Czech J. Tourism* 3 (1), 5–26.
- Dyer, P., Gursoy, D., Sharma, B., Carter, J., 2007. Structural modeling of resident perceptions of tourism and associated development on the Sunshine Coast, Australia. *Tour. Manage.* 28 (2), 409–422.
- Farber, S., O'Kelly, M., Miller, H.J., Neutens, T., 2015. Measuring segregation using patterns of daily travel behavior: a social interaction based model of exposure. *J. Transp. Geogr.* 49, 26–38.
- Fan, D.X.F., Zhang, H.Q., Jenkins, C.L., Tavitiyaman, P., 2017. Tourist typology in social contact: an addition to existing theories. *Tour. Manag.* 60, 357–366.
- Golledge, R., Stimson, R., 1997. *Spatial Behavior: a Geographic Perspective.* The Guilford Press, New York.
- Hägerstrand, T., 1970. What about people in regional science?. In: *Papers of the Regional Science Association*, pp. 7–21.
- James, D.R., Taeuber, K.E., 1985. Measures of segregation. *Sociological methodology* 15, 1–32.
- Janelle, D. G. (Ed.), 1995. *Metropolitan expansion, telecommuting and transportation. The geography of urban transportation.* Guilford, New York, pp. 407 – 34.
- Jurowski, C., Gursoy, D., 2004. Distance effects on RESIDENTS' ATTITUDES toward tourism. *Ann. Tour. Res.* 31 (2), 296–312.
- Krippendorf, J., 1987. Ecological approach to tourism marketing. *Tour. Manag.* 8 (2), 174–176.
- Kwan, M.P., Schwanen, T., 2016. Geographies of mobility. *Ann. Am. Assoc. Geogr.* 106 (2), 243–256.
- Li, D., Zhou, X., Wang, M., 2018. Analyzing and visualizing the spatial interactions between tourists and locals: a Flickr study in ten US cities. *Cities* 74, 249–258.
- Lukasová, A., 1979. Hierarchical agglomerative clustering procedure. *Pattern Recognition* 11 (5–6), 365–381.
- Lundberg, D.E., 1974. Caribbean tourism. *Cornell Hotel Restaur. Admin. Q.* 14 (4), 30–45.
- Liu, Y., Wang, S., Wang, X., Zheng, Y., Chen, X., Xu, Y., Kang, C., 2024. Towards semantic enrichment for spatial interactions. *Ann. GIS* 30 (2), 151–166.
- Liu, Y., Liu, X., Gao, S., Gong, L., Kang, C., Zhi, Y., Chi, G., Shi, L., 2015. Social sensing: a new approach to understanding our socioeconomic environments. *Ann. Assoc. Am. Geogr.* 105 (3), 512–530.
- Li, M., Gao, S., Lu, F., Liu, K., Zhang, H., Tu, W., 2021. Prediction of human activity intensity using the interactions in physical and social spaces through graph convolutional networks. *Int. J. Geogr. Inf. Sci.* 35 (12), 2489–2516.
- Mason, P., Cheyne, J., 2000. Residents' attitudes to proposed tourism development. *Ann. Tour. Res.* 27 (2), 391–411.
- Mashkov, R., Shoval, N., 2023. Using high-resolution GPS data to create a tourism Intensity-Density Index. *Tour. Geogr.* 25 (6), 1657–1678.
- Massey, D.S., Denton, N.A., 1988. The dimensions of residential segregation. *Soc. Forces* 67, 281–315.
- Mcelroy, J.L., 2003. Tourism development in small islands across the world. *Geografiska Annaler: Series b, Human Geogr.* 85 (4), 231–242.
- Miller, H., 2004. Activities in space and time. In: Hensher, D., Button, K., Haynes, K., Stopher, P. (Eds.), *Handbook of Transport 5: Transport Geography and Spatial Systems.* Elsevier Science, London, pp. 647–660.
- Miller, H.J., 2005a. Necessary space-time conditions for human interaction. *Environ. Plann. b: Plann. Design* 32 (3), 381–401.
- Miller, H.J., 2005b. A measurement theory for time geography. *Geogr. Anal.* 37 (1), 17–45.
- Mody, M., Hanks, L., Dogru, T., 2019. Parallel pathways to brand loyalty: Mapping the consequences of authentic consumption experiences for hotels and Airbnb. *Tour. Manag.* 74, 65–80.
- Neutens, T., Schwanen, T., Witlox, F., 2011. The prism of everyday life: Towards a new research agenda for time geography. *Transp. Rev.* 31 (1), 25–47.
- Parkes, D., Thrift, N., 1980. *Times, Spaces, and Places: a Chronogeographic Perspective.* John Wiley, New York.
- Pizam, A., Uriely, N., Reichel, A., 2000. The intensity of tourist–host social relationship and its effects on satisfaction and change of attitudes: the case of working tourists in Israel. *Tour. Manag.* 21 (4), 395–406.
- Raymond, C., Brown, G., 2007. A spatial method for assessing resident and visitor attitudes towards tourism growth and development. *J. Sustain. Tour.* 15 (5), 520–540.
- Sharpley, R., 2014. Host perceptions of tourism: a review of the research. *Tour. Manag.* 42, 37–49.
- Shaw, S.L., Sui, D., 2020. Understanding the new human dynamics in smart spaces and places: toward a spatial framework. *Ann. Am. Assoc. Geogr.* 110 (2), 339–348.
- Shaw, S.L., Yu, H., 2009. A GIS-based time-geographic approach of studying individual activities and interactions in a hybrid physical–virtual space. *J. Transp. Geogr.* 17 (2), 141–149.
- Shaw, S.L., 2012. Guest editorial introduction: time geography—its past, present and future. *J. Transp. Geogr.* 23, 1–4.
- Sherlock, K., 2001. Revisiting the concept of hosts and guests. *Tour. Stud.* 1 (3), 271–295.
- Ren, M., Xu, Y., Park, S., Huang, X., Sun, M., Xia, J., Koh, S.Y., 2024. Consumer spending during COVID-19 in a tourism city. *Annals of Tourism Research* 109, 103830.
- Sharma, B., Dyer, P., 2009. An investigation of differences in residents' perceptions on the Sunshine Coast: tourism impacts and demographic variables. *Tour. Geogr.* 11 (2), 187–213.
- Sheldon, P.J., Var, T., 1984. Resident attitudes to tourism in North Wales. *Tour. Manag.* 5 (1), 40–47.
- Shoval, N., Isaacson, M., 2007. Sequence alignment as a method for human activity analysis in space and time. *Ann. Assoc. Am. Geogr.* 97 (2), 282–297.
- Snepenger, D.J., Murphy, L., O'Connell, R., Gregg, E., 2003. Tourists and residents use of a shopping space. *Ann. Tour. Res.* 30 (3), 567–580.
- Su, X., Hooimeijer, P., Spierings, B., 2021. Why urban setting matters in shaping tourist attitudes towards interaction with residents: Causation or selection in three urban settings. *J. Destin. Mark. Manag.* 22, 100657.
- Su, X., Spierings, B., Hooimeijer, P., 2022. Different urban settings affect multi-dimensional tourist-resident interactions. *Tour. Geogr.* 24 (4–5), 815–836.
- Sui, D., 2012. Looking through Hägerstrand's dual vistas: towards a unifying framework for time geography. *J. Transp. Geogr.* 23, 5–16.
- Tan, G., Hsu, C.H., 2024. Can host–tourist interaction pull the plug on tourist stereotypes? Exploring the power of emotional solidarity. *Tour. Manag. Perspect.* 50, 101211.
- Teye, V., Sirakaya, E., Sönmez, S.F., 2002. Residents' attitudes toward tourism development. *Ann. Tour. Res.* 29 (3), 668–688.
- Tosun, C., 2002. Host perceptions of impacts: a comparative tourism study. *Ann. Tour. Res.* 29 (1), 231–253.
- UNESCO, (2024). *Travel without leaving a trace. The UNESCO Courier.* <https://courier.unesco.org/en/articles/travel-without-leaving-trace>.
- Victoriano, R., Paez, A., Carrasco, J.A., 2020. Time, space, money, and social interaction: using machine learning to classify people's mobility strategies through four key dimensions. *Travel Behav. Soc.* 20, 1–11.
- Vu, H.Q., Muskat, B., Li, G., Law, R., 2021. Improving the resident-tourist relationship in urban hotspots. *J. Sustain. Tour.* 29 (4), 595–615.
- Wall, G., Mathieson, A., 2006. *Tourism: Change, Impacts, and Opportunities.* Pearson Education.
- Weaver, D.B., Lawton, L.J., 2001. Resident perceptions in the urban-rural fringe. *Ann. Tour. Res.* 28 (2), 439–458.
- Wong, D.W., 2002. Modeling local segregation: a spatial interaction approach. *Geogr. Environ. Model.* 6 (1), 81–97.
- Xu, Y., Shaw, S.L., Zhao, Z., Yin, L., Fang, Z., Li, Q., 2015. Understanding aggregate human mobility patterns using passive mobile phone location data: a home-based approach. *Transportation* 42, 625–646.
- Xu, Y., Belyi, A., Bojic, I., Ratti, C., 2017. How friends share urban space: an exploratory spatiotemporal analysis using mobile phone data. *Trans. GIS* 21 (3), 468–487.
- Xu, Y., Belyi, A., Santi, P., Ratti, C., 2019. Quantifying segregation in an integrated urban physical-social space. *J. R. Soc. Interface* 16 (160), 20190536.
- Yang, X., Fang, Z., Xu, Y., Shaw, S.L., Zhao, Z., Yin, L., Zhang, T., Lin, Y., 2016. Understanding spatiotemporal patterns of human convergence and divergence using mobile phone location data. *ISPRS Int. J. Geo Inf.* 5 (10), 177.
- Yin, L., Shaw, S.L., 2015. Exploring space-time paths in physical and social closeness spaces: a space-time GIS approach. *Int. J. Geogr. Inf. Sci.* 29 (5), 742–761.
- Zhang, G., Luo, W., Wu, M., Ye, L., 2025. Exploring social interaction patterns and drivers in VGI communities using a custom geovisual analytics tool. *Ann. GIS* 31 (3), 413–431.
- Zhou, J., Yang, Y., Ma, H., Li, Y., 2020. "Familiar strangers" in the big data era: an exploratory study of Beijing metro encounters. *Cities* 97, 102495.